

Article

Climate Adaptation Measures for Enhancing Urban Resilience

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Abstract: Climate change threatens urban areas globally. Enhancing resilience is crucial, yet the comprehensive clustering of practical climate adaptation measures for use in construction industry decision-making is notably absent. This study investigates and categorizes various climate adaptation measures, justifying each through a lens of risk management, asset management, and previous scientific work. It takes advantage of the innovative digital platform Netobra, which offers an ecosystem for the construction industry, to provide real-world, practical implications of these measures. Informed by the Urban Resilience Evaluation System, ISO 31000 (risk management), and ISO 55000 (asset management), the study sets out to demonstrate the value of these measures in bolstering urban resilience and improving decision-making in the construction industry. Moreover, the study integrates a hotspot detection mechanism for areas at high risk of climate impacts, using multicriteria decision analysis (MCDA)—analytic hierarchy process (AHP) mapping on Netobra. These identified hotspots and corresponding climate adaptation measures will further be incorporated into a Risk-Informed Asset-Centric (RIACT) process, providing valuable insights for climate change mitigation and adaptation strategies in urban development. Through its in-depth analysis, the study aims to contribute to the understanding of how diverse climate adaptation measures can be practically applied in various sectors, thereby enhancing urban resilience and effective risk and asset management.



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Keywords: climate adaptation measures; urban resilience; risk management (ISO 31000); asset management (ISO 55000); multicriteria decision analysis (MCDA)

1. Introduction

Climate change, characterized by rising global temperatures, the increasing frequency of extreme weather events, and sea-level rise, has significant implications for cities around the world [1]. Urban areas, with their high concentration of people, infrastructure, and economic activity, are particularly vulnerable to these impacts. Urban resilience, or the ability of urban systems to withstand and recover from shocks and stresses, is a critical concept in addressing these challenges.

As part of efforts to build urban resilience, there is an increasing recognition of the need for effective climate adaptation measures [2]. These measures, which involve adjustments in ecological, social, or economic systems in response to actual or expected climate stimuli, play a crucial role in reducing the vulnerability of urban areas and ensuring their continued functioning in the face of climate change.

This paper aims to explore and categorize various climate adaptation measures, justifying each based on previous work, risk management, and asset management practices. The study will use the Netobra.com online platform, which serves as an ecosystem for the construction industry, as a case study for the practical application of these measures.

The analysis provided in this paper will enhance our understanding of how diverse climate adaptation measures can be integrated into urban resilience strategies [3]. Moreover, it will demonstrate how these strategies can be operationalized within the construction

industry, a sector of key importance in urban development, to improve decision-making and risk management.

Our study will be underpinned by the Urban Resilience Evaluation System (URES) [4]. This system uses various dimensions, indicators, and parameters to assess and track the resilience of urban areas to climate impacts. This evaluation system provides a systematic framework for understanding the multifaceted nature of urban resilience and identifying areas for improvement [5].

The ISO 31000 [6] and ISO 55000 [7] standards serve as guiding principles in our selection and application of the 50 climate adaptation measures. The ISO 31000 principles provide a robust framework for managing risks, aiding in the identification and prioritization of risks associated with climate change. This risk management approach directly influences the selection of adaptation measures. On the other hand, the ISO 55000 principles guide the management of assets, which are crucial for the implementation of the adaptation measures. By integrating these ISO standards, we ensure that our climate adaptation strategies are not only effective in mitigating climate risks but also sustainable in managing our resources [8].

One of the key applications of our study's findings is their integration into Netobra.com, a newly developed innovative online platform. This platform provides a collaborative space for professionals in the construction industry, facilitating knowledge exchange, resource optimization, and better project management. Through this integration, we aim to enhance the industry's capacity to implement climate adaptation measures and build urban resilience [9].

Furthermore, our study will incorporate an analysis of disaster risk hotspots using multicriteria decision analysis (MCDA)—analytic hierarchy process (AHP) mapping on Netobra.com. This analysis will enable us to identify areas at high risk of climate impacts and prioritize adaptation measures accordingly [10].

The structure of this paper is designed to provide a thorough examination of the different facets of climate adaptation measures, particularly their role in enhancing urban resilience [11]. The paper begins with an introduction and a brief overview, followed by a statement of the study's purpose and objectives. It then presents the theoretical framework that underpins the research and explains the scope and application of the study. This is followed by a clear articulation of the paper's thesis statement.

In its main body, the paper first offers a systematic review of climate adaptation measures, including an analysis of research trends, the identification of research gaps, and conclusions drawn from the systematic review. This is followed by a detailed review of various types of climate adaptation measures and their roles in strengthening urban resilience [11]. This portion of the paper examines physical, ecosystem, water management, policy, and health strategies in climate adaptation. This review is complemented by a discussion on how climate adaptation measures can be integrated with risk and asset management for enhanced urban resilience. In the subsequent sections, the paper presents an application case of these measures on Netobra.com, including the role of this platform in the construction industry and its use of multicriteria decision analysis (MCDA) and risk impact assessment and control (RIACT) processes [12]. The paper concludes with a section on the results and discussion, encapsulating the key findings and their interpretation and implications for urban resilience and the construction industry. The conclusion then recapitulates the findings, explains the contribution to the field, and ends with suggestions for future research.

The central argument of this paper is that understanding and implementing a diverse range of climate adaptation measures, across various sectors, is essential for enhancing urban resilience. These measures can improve risk management and asset management practices and enable effective decision-making in construction [13].

2. Systematic Review of Climate Adaptation Measures

The importance of addressing climate change adaptation measures has been increasingly recognized within academic and policy circles over the past decades [14]. As a consequence, there has been a surge in the literature on this subject, making it crucial to provide a systematic review of this rapidly growing field. The primary objective of our study is to explore the knowledge evolution and research trends regarding climate change adaptation measures by performing a systematic literature review. By scrutinizing 661 documents selected based on specified criteria, we aim to map the field, explore the subject areas involved, and analyze the publishing trend over the years [15].

We chose Scopus as the primary database for our systematic review due to its extensive coverage of literature across a wide range of disciplines, its user-friendly interface, and its capabilities for citation analysis. Scopus indexes a larger number of journals and conference papers compared to Web of Science (WoS), which increased the likelihood of identifying a larger number of relevant articles for our review. Furthermore, Scopus' inclusive indexing of different document types, including journal articles, conference papers, and patents, provided a broader scope for our review. While we recognize the value of other databases such as WoS, we ensured a comprehensive and inclusive literature review by cross verifying the covered literature with other databases.

2.1. Scopus Advance Search String Selection

The search string for this research was carefully formulated considering various factors to ensure the extraction of the most relevant literature. The chosen search string is TITLE-ABS-KEY (climate AND adaptation AND measures) AND (LIMIT-TO (EXACTKEYWORD, "Climate Change Adaptation")) AND (LIMIT-TO (PUBSTAGE, "final")) AND (LIMIT-TO (SRCTYPE, "j")). The rationale for each element of this search string is as follows: the TITLE-ABS-KEY limits the search to those documents where the keywords appear in the title, abstract, or keyword section, ensuring relevancy. The LIMIT-TO function was employed to narrow down the search to those with the exact keyword "Climate Change Adaptation" to focus on the chosen theme. Limiting the PUBSTAGE to "final" ensures that only completed articles are considered, excluding preliminary or in-process works. The final limitation to "j" in SRCTYPE ensures that only journal articles are considered, given their rigorous peer-review process and generally their high quality.

2.2. Analysis of the Year

Our analysis of the distribution of documents over the years revealed an interesting trend. From 1996 until 2007, there were only minimal publications in this field, reflecting the nascent stage of climate change adaptation research. A gradual increase could be seen from 2007, which aligns with the growing global realization of climate change. A noticeable surge in the number of published documents started in 2010, with the highest publications reported in 2021 (87 documents) followed by 2022 (80 documents) and 2023 (34 documents in the first six months of the year). This escalating trend signifies the increasing importance given to climate change adaptation measures in academic research; See Figure 1.

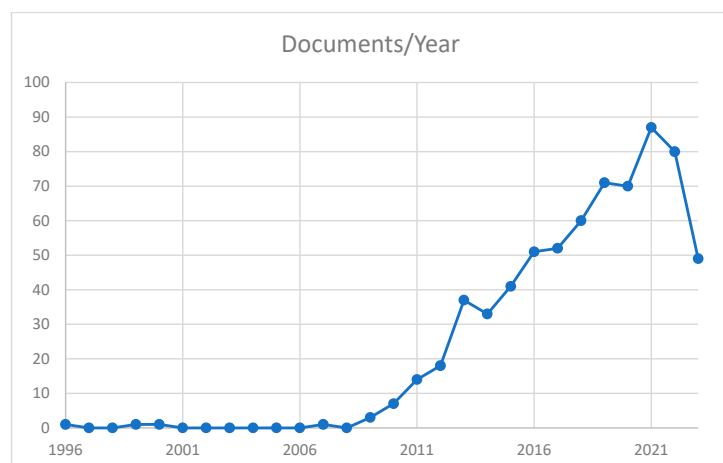


Figure 1. Document distribution over the years of the global trend in climate adaptation publications from 1996 to 2023, reflecting increased awareness and attention starting around 2010.

2.3. Analysis of Subject Area

In analyzing the subject area of the documents, Environmental Science dominates with 491 documents, reflecting the direct impact of climate change on the environment. Social Sciences, with 240 documents, and Earth and Planetary Sciences, with 163 documents, also contribute significantly. It is encouraging to see a broad interdisciplinary interest, with contributions from Engineering, Agricultural and Biological Sciences, Energy, and Economics. The diversity of the subject areas involved underlines the complex and interdisciplinary nature of climate change adaptation measures [16]; See Figure 2.

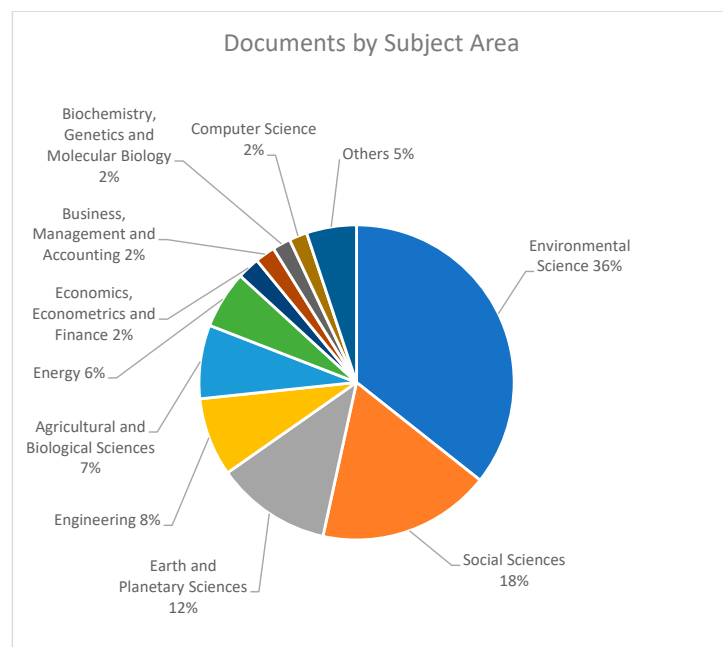


Figure 2. Analysis of subject areas in research that breaks down climate adaptation research subjects, highlighting Environmental Science, Social Sciences, Earth and Planetary Sciences, and other interdisciplinary fields.

2.4. Research Trend Analysis

The body of literature on climate change adaptation measures has evolved significantly over the past decade, indicating an increased recognition of the complexity and urgency of the issue [17]. In our systematic review, we identified several key themes that are

consistently emphasized in the current literature. These themes, along with representative papers, are presented in Table 1.

Table 1. Key themes identified in the systematic review of climate adaptation measures for urban resilience, with representative papers.

Theme	Example Papers
Emphasis on cross-sector collaboration	“Truths of the riverscape: Moving beyond command-and-control to geomorphologically informed nature-based river management” [18], “Cross-scale collaboration for adaptation to climate change: a two-mode network analysis of bridging actors in Switzerland” [19]
Increased focus on local adaptations	“Climate change adaptation in smallholder agriculture: adoption, barriers, determinants, and policy implications” [20], “Adaptations of market garden producers to climate change in southern Mali” [21]
Integration of technology and artificial intelligence	“Combined artificial intelligence, sustainable land management, and stakeholder engagement for integrated landscape management in Mediterranean watersheds” [22]
Consideration of socio-economic factors	“Empowering the voiceless: Securing the participation of marginalized groups in climate change governance in South Africa” [23], “Determinants, outcomes, and feedbacks associated with microeconomic adaptation to climate change” [24]
Inclusion of nature-based solutions	“Tamm review: Ecological principles to guide post-fire forest landscape management in the Inland Pacific and Northern Rocky Mountain regions” [25]

As summarized in Table 1, these themes represent the breadth of approaches and perspectives in the current discourse on climate adaptation measures for urban resilience.

2.5. Research Gap Identification

Despite the growing body of research, some gaps persist in the literature on climate change adaptation measures:

- (i) Limited research on the effectiveness of adaptation measures: Few papers, such as “Assessing climate change mitigation and adaptation strategies and agricultural innovation systems in the Niger Delta” [26], thoroughly evaluate the effectiveness of implemented measures, indicating a need for more rigorous assessment.
- (ii) Absence of long-term studies: Most studies are cross-sectional, while longitudinal studies that assess the sustained impact of adaptation measures over time are rare [27,28].
- (iii) Unequal geographic representation: There is a skewed representation towards certain geographical areas, while other regions, especially developing countries, are under-represented [29,30].
- (iv) Lack of focus on cultural aspects: Cultural perspectives on climate change adaptation are not adequately discussed in the literature, which could offer unique insights into community-level adaptation measures [31].
- (v) Minimal consideration of indigenous knowledge: Despite increasing recognition of its value, indigenous knowledge is not sufficiently incorporated into climate change adaptation research [32,33].

In summary, this systematic literature review has delved deeply into the multidimensional field of climate change adaptation measures, shedding light on its evolution and current trends [34,35]. Our examination has demonstrated an intensifying trend in research output, a testament to the mounting global urgency and commitment to address climate change adaptation. The multitude of involved subject areas echoes the multifaceted nature of climate adaptation, thus necessitating a broad and interdisciplinary approach [36,37]. This review further exposes critical gaps in the existing body of literature, including the need for comprehensive effectiveness evaluations, long-term impact studies,

better geographical representation, and more robust integration of cultural and indigenous perspectives [38]. Consequently, this comprehensive review sets the stage for more refined future research and policy discussions, with the goal of promoting comprehensive, sustainable, and context-specific climate change adaptation strategies [39,40].

2.6. Review of Climate Adaptation Measures

Urban resilience goes beyond merely withstanding climate-related shocks; it also encompasses the capacity of cities to adapt and transform [41,42]. Climate adaptation measures serve as essential tools in this process, bolstering physical infrastructure, strengthening social systems, optimizing water resources, and informing policies. By implementing such measures, from climate-sensitive building codes to water-saving strategies, cities can enhance their resilience and navigate the complexities of climate change [43,44].

Climate change poses profound challenges for societies globally, necessitating the development and implementation of diverse, multifaceted climate adaptation measures to mitigate impacts and promote resilience [45]. These strategies span five broad categories: physical infrastructure [46,47], ecosystem restoration and protection [48,49], water management [50,51], and policy and planning [52,53]. Each provides unique responses to different aspects of climate change, and together, they present a comprehensive framework for understanding the range of adaptation strategies available.

Understanding Climate Adaptation Measures: Climate adaptation measures are strategies implemented to counteract the adverse effects of climate change while capitalizing on potential beneficial opportunities. These measures encompass physical infrastructure changes, such as sea walls to protect against rising sea levels, and disease monitoring programs to mitigate climate-induced health risks. Additionally, strategic urban planning that factors in climate change forms a significant part of these measures. Although varied in nature, these measures share the collective objective of boosting resilience to climate change in both natural and human-made systems [54].

Role of Climate Adaptation Measures in Urban Resilience: Urban resilience embodies the capacity of cities to survive, adapt, and grow, regardless of the chronic stresses and acute shocks they experience. As urban areas are particularly vulnerable to climate change impacts, climate adaptation measures become crucial tools for these areas to proactively manage and mitigate these threats. For instance, by enforcing climate-sensitive building codes, cities can enhance their built environments' resilience. Moreover, by adopting water-saving strategies and drought-tolerant crops, cities can protect their water and food supplies, further enhancing their resilience.

2.7. Review of Physical, Ecosystem, Water Management, Policy, and Health Strategies in Climate Adaptation

Physical Infrastructure: The construction and retrofitting of physical infrastructure to better endure the challenges posed by climate change is a significant aspect of climate adaptation. Examples of these measures include sea walls and flood barriers [55], which have been implemented in various coastal regions as frontline defenses against sea-level rise and flooding. Likewise, upgrading stormwater and sewage systems to handle increased rainfall events [56], improving building insulation to reduce heatwave impacts [57–59], and exploring heat-resistant pavement materials are other key areas [60]. Importantly, fire-resistant materials and designs for homes have gained attention, particularly in fire-prone regions [61,62].

Ecosystem Restoration and Protection: Protecting and restoring ecosystems are crucial strategies for climate adaptation [63,64]. These nature-based solutions offer numerous co-benefits, including carbon sequestration, flood reduction, and urban cooling [65,66]. Reforestation efforts, for example, have both local and global benefits, including climate regulation and carbon storage [67]. Wetlands can play a vital role in absorbing flood waters, and the protective function of coastal ecosystems such as mangroves is well-recognized [68]. Urban tree planting can offer localized cooling effects and improve urban

living conditions [69,70]. Moreover, conserving genetic diversity in crops and livestock is seen as a key strategy for ensuring food security under changing climatic conditions [71].

Water Management: Climate change has significant implications for water availability, necessitating a range of water management strategies [72]. Rainwater harvesting systems can increase water security, especially in areas experiencing irregular rainfall, while the development of drought-tolerant crops can help ensure food security [73,74]. Greywater recycling can reduce freshwater demand, and efficient household water use strategies can contribute to overall water conservation [75]. Additionally, protecting groundwater resources is a critical aspect of water management under climate change [76,77].

Policy and Planning: Policy and planning are critical components of climate adaptation, enabling the implementation of both direct measures and facilitating others [78]. For example, policies that support renewable energy can mitigate climate change by reducing greenhouse gas emissions [79], while tax incentives can encourage the adoption of climate-resilient practices [80]. Education programs can raise awareness and equip communities with the necessary knowledge and tools to respond effectively to climate change [81]. Further, considering the possibility that some areas may become uninhabitable due to extreme climate impacts, climate migration policies could become necessary [82].

Health and Social Measures: Adapting to climate change necessitates comprehensive measures that address not just physical or environmental factors, but also the health and social dimensions [83]. An array of strategies is imperative for the wellbeing of communities in the face of the emerging challenges. Disease monitoring and prevention programs, heat-health action plans, enhanced emergency services, food security programs, and early warning systems form an integral part of this holistic approach [84–86]. These strategies aim to mitigate the adverse health impacts that might arise due to climate change, such as the rise of vector-borne diseases and increased mortality during heatwaves, and ensure a rapid and efficient response to climate-induced disasters [87].

In recognizing the psychological strain that climate change and related disasters can impose on individuals, the role of mental health support cannot be understated [88,89]. Community-based adaptation initiatives leverage local knowledge and participation for resilient outcomes, ensuring that adaptation strategies are culturally appropriate and accepted [90]. The provision of cooling centers for vulnerable populations offers immediate relief during heatwaves, while training programs equip communities with the necessary skills to navigate climate impacts [91]. Thus, these health and social measures underscore the human-centered approach required for effective climate resilience and adaptation [92,93].

This review underscores the diverse array of climate adaptation measures available, ranging from physical infrastructure improvements to policy changes. It is evident that no single measure is a silver bullet; rather, a multifaceted, context-specific suite of measures will be necessary to ensure resilience in the face of climate change [94]. These measures, when strategically implemented, can help safeguard communities, economies, and ecosystems from the adverse impacts of climate change [95].

Integrating Climate Adaptation Measures and Risk and Asset Management for Enhanced Urban Resilience: Urban areas globally are recognizing the need for comprehensive climate adaptation measures to enhance resilience amidst escalating climate change impacts. Such measures not only encompass physical infrastructure alterations and ecosystem-based strategies but also rely on the integration of robust risk and asset management principles [96].

Several cities have already begun demonstrating the effectiveness of such climate adaptation strategies. For example, Rotterdam has developed an extensive network of green roofs and water plazas, countering the city's increasing risk of flooding due to climate change [97]. Similarly, Melbourne's Urban Forest Strategy, which aims to double tree cover by 2040, is mitigating the urban heat island effect and enhancing the city's resilience to rising temperatures. Singapore is also stepping up its climate adaptation efforts, embarking on a coastal protection study to safeguard the low-lying city-state from sea-level rise. These examples reflect the variety of climate adaptation measures that urban areas can utilize to fortify their resilience.

As part of these adaptation strategies, urban areas are increasingly integrating principles of risk and asset management, primarily through the frameworks provided by ISO 31000 and ISO 55000 [6,7,44]. ISO 31000, the international standard for risk management, emphasizes the systematic, transparent, and reliable process of identifying, understanding, and managing risk [98]. In contrast, ISO 55000, the international standard for asset management, focuses on maximizing the value derived from assets throughout their lifecycle [99]. The integration of these two standards can substantially augment the effectiveness of climate adaptation measures [44,97].

By infusing the systematic approach of ISO 31000 into climate adaptation, cities can identify and assess climate-related risks more efficiently and develop effective mitigation and adaptation measures. Concurrently, utilizing ISO 55000's emphasis on lifecycle asset management ensures that urban infrastructure assets, such as buildings, roads, and utilities, are resilient to climatic stresses and continue to deliver value [100]. For instance, cities might prioritize upgrades to stormwater and sewage systems as a climate adaptation measure to ensure these systems can withstand increased rainfall due to climate change [44].

The integration of ISO 31000 and ISO 55000 principles into climate adaptation strategies has far-reaching implications for urban resilience [99]. It not only promotes a more proactive and systematic approach to managing climate risks but also maximizes the value derived from urban assets, despite climatic challenges [101]. As cities continue to grow and face increasing climate change impacts, the integration of these principles into climate adaptation strategies significantly contributes to their resilience, safeguarding people, infrastructure, and economic activity against climatic threats [102]. In this way, the marriage of robust climate adaptation measures and well-structured risk and asset management principles offers a promising pathway to enhanced urban resilience in the era of climate change [96].

3. Clustering of Climate Adaptation Measures

Climate adaptation measures constitute a variety of strategies designed to augment urban resilience. These strategies are vast and diverse, reflecting the multifaceted nature of climate change and its impacts. To provide a comprehensive yet manageable overview, these measures can be grouped into five main dimensions: physical infrastructure, ecosystem restoration and protection, water management, policy and planning, and health and social considerations.

Each dimension represents a crucial aspect of climate adaptation, illustrating the interconnectedness of various sectors in addressing the challenges of climate change. Physical infrastructure focuses on strengthening built environments, while ecosystem restoration and protection involve the conserving and reviving of natural ecosystems. Water management strategies ensure efficient use and conservation of water resources. Policy and planning refer to regulatory and institutional frameworks guiding adaptation strategies. Lastly, health and social measures aim to safeguard public health and societal structures from climate impacts. This classification not only brings clarity to the vast array of adaptation measures but also underscores the necessity of a multisectoral approach in building urban resilience.

Figure 3 provides a visual representation of these categories, illustrating the interconnectedness and importance of each dimension in addressing the challenges posed by climate change.

The choice to identify and classify climate adaptation measures into five main categories, each containing ten subcategories, was motivated by several key factors.

First, it was essential to maintain a balance between comprehensiveness and manageability. While there are numerous potential measures to counter climate change impacts, incorporating too many can lead to an overly complex and unwieldy framework, making it difficult for stakeholders to comprehend and effectively implement. On the other hand, a list that is too short could overlook important measures, leading to an insufficiently robust response to climate change.

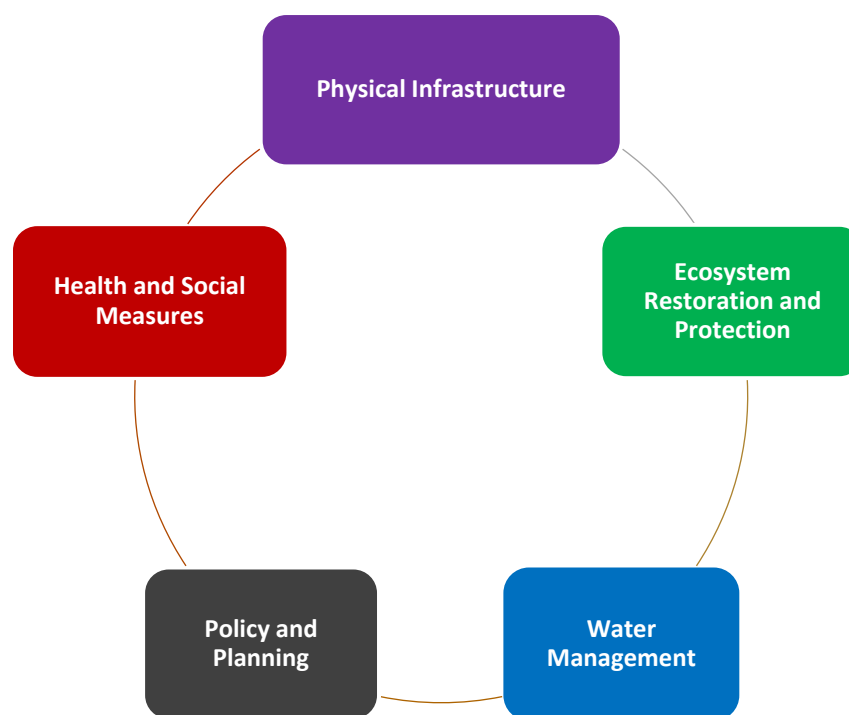


Figure 3. Dimensions of climate adaptation as the importance of interconnected dimensions—physical infrastructure, ecosystem restoration, water management, policy and planning, and health and social considerations—in enhancing urban resilience.

Ten subcategories within each of the five main categories provide a good balance. This number allows for broad coverage of the diverse range of measures available, without overwhelming the user with an excessively detailed list. Each main category encapsulates a distinct area of climate adaptation, ensuring that the full breadth of possible approaches is covered.

Second, the five main categories were chosen to reflect the multifaceted nature of the climate change challenge, which requires a multisectoral response—involving physical infrastructure, ecosystem restoration, water management, policy planning, and health and social measures. This categorization underscores the interconnected nature of climate adaptation measures, indicating that no single area can address climate change in isolation.

Lastly, the ten subcategories within each main category allow for a degree of granularity that makes the measures more actionable. Rather than merely referring to broad strategies, this structure breaks down each category into specific tactics, making it easier for stakeholders to understand what is needed and how to implement these measures. Each subcategory is a realistic, attainable action that cities, organizations, and individuals can take to build resilience against climate change.

3.1. Physical Infrastructure

The physical infrastructure of communities is a crucial line of defense against the impacts of climate change. The adaptation measures under this category are concerned with enhancing the resilience of the built environment to climate extremes, primarily focused on combating the effects of rising temperatures, altered precipitation patterns, rising sea levels, and an increased frequency of extreme weather events [103,104].

1. Sea walls to protect against sea-level rise: Sea walls have been traditionally used as a frontline defense against coastal erosion and flooding. With the anticipated sea-level rise due to climate change, the importance of sea walls is even more pertinent [55,105].

2. Flood barriers and levees: Similarly, flood barriers and levees play a significant role in flood-prone areas, particularly in regions where climate models predict increased precipitation [106,107].
3. Drought-resistant agricultural practices: Climate change impacts agricultural productivity by altering precipitation patterns and increasing temperatures. Adaptation measures such as drought-resistant crop varieties and efficient irrigation systems have been suggested to enhance resilience in the agriculture sector [108,109].
4. Upgrading stormwater and sewage systems: As climate change is expected to increase the frequency of extreme weather events, upgrading stormwater and sewage systems can prevent flooding and related public health issues [110].
5. Improving building insulation: Improved insulation in buildings can contribute to energy efficiency and thermal comfort, especially in regions experiencing warmer temperatures [57,111].
6. Heat-resistant pavement materials: Heat-resistant pavements can reduce the urban heat island effect and increase the lifespan of road infrastructures in hot climates [112].
7. Construction of cooling centers: Cooling centers provide safe, air-conditioned environments during heatwaves, protecting vulnerable populations from heat stress [113,114].
8. Green roofs and walls: Green roofs and walls, along with urban forests, offer multiple benefits, including reduced energy demand for cooling, urban heat island mitigation, stormwater runoff reduction, and increased urban biodiversity [115,116].
9. Development of floating or stilt-based structures: These structures can offer a viable solution for communities that face increased flooding or sea-level rise. Floating homes are already in use in parts of the Netherlands [117].
10. Fire-resistant materials and designs for homes: This is especially relevant in regions prone to wildfires. Fire-resistant building designs and materials can significantly reduce the vulnerability of houses to fire damage [118,119].

Thus, the adaptation of physical infrastructure forms an integral part of climate change resilience, protecting communities from imminent climatic changes while also reducing the environmental impact of the built environment.

3.2. Ecosystem Restoration and Protection

Ecosystem restoration and protection, often referred to as nature-based solutions, play a pivotal role in climate adaptation by enhancing natural resilience and promoting biodiversity. These measures not only combat the effects of climate change but also provide numerous co-benefits, such as improved air quality, carbon sequestration, and enhanced habitats for wildlife.

1. Reforestation and afforestation: Forests act as major carbon sinks, contributing to the mitigation of greenhouse gas emissions. They also regulate local climate, reduce erosion, and support biodiversity. Reforestation and afforestation are crucial for restoring these functions [120].
2. Wetland restoration to absorb floodwaters: Wetlands have a natural capacity to absorb and store floodwaters, providing a buffer against sea-level rise and storms. Restoration of these ecosystems can reduce the vulnerability of nearby communities to flooding [121,122].
3. Coral reef restoration for coastal protection: Healthy coral reefs form barriers against wave action, thus protecting coastlines from erosion. As climate change threatens these ecosystems, their restoration becomes an important adaptation strategy [123].
4. Protection and expansion of mangrove forests: Mangrove forests provide significant coastal protection and sequester large amounts of carbon. Protecting and expanding these ecosystems can offer a dual benefit of adaptation and mitigation [124–126].
5. Soil conservation practices: Healthy soils are crucial for food production and carbon storage. Practices such as cover cropping, reduced tillage, and organic amendments can enhance soil health and resilience to climatic stresses [127].

6. Peatland restoration and conservation: Peatlands store a significant portion of the world's soil carbon. Their degradation releases this carbon into the atmosphere, while their restoration and conservation can maintain these carbon stocks and support unique ecosystems [128,129].
7. Establishment of wildlife corridors: Climate change can shift the ranges of many species, and wildlife corridors can facilitate these movements, promoting species survival and ecosystem resilience [130].
8. Planting urban trees for shade and cooling: Urban trees provide shade and evapotranspiration, reducing the urban heat island effect. They also improve air quality and urban aesthetics [131,132].
9. Invasive species management: Climate change can facilitate the spread of invasive species, which can outcompete native species and disrupt ecosystems. Invasive species management can protect biodiversity and ecosystem function [133].
10. Conservation of genetic diversity in crops and livestock: Genetic diversity can provide a buffer against climate change by allowing species to adapt to changing conditions. This is particularly important in agriculture, where crop and livestock diversity can support food security in the face of climate change [71].

Finally, ecosystem restoration and protection, by maintaining and enhancing the services provided by natural ecosystems, form a crucial part of a holistic approach to climate adaptation.

3.3. Water Management

Water management plays a pivotal role in climate change adaptation. As climate change alters the hydrological cycle, resulting in extreme events such as droughts and floods, efficient water management can significantly mitigate these impacts.

1. Rainwater harvesting systems: Rainwater harvesting can provide a valuable supplementary water source during periods of low rainfall or drought, reducing the reliance on conventional water sources [134,135].
2. Development of drought-tolerant crops: In regions affected by increased drought frequency, the use of drought-tolerant crop varieties can sustain agricultural productivity and contribute to food security [136].
3. Greywater recycling systems: Recycling greywater for non-potable uses reduces the demand for freshwater resources, which is crucial in water-scarce regions [75].
4. Advanced irrigation techniques (e.g., drip irrigation): Efficient irrigation systems can significantly reduce water use in agriculture while maintaining or even increasing crop yields [137,138].
5. Desalination plants: In coastal areas, desalination can provide a reliable water supply irrespective of climate variability, although energy demands and environmental impacts should be carefully managed [139,140].
6. Protection of groundwater resources: Groundwater can provide a buffer against drought, but over-extraction can lead to resource depletion and other problems. Protection and sustainable management of groundwater are therefore crucial adaptation strategies [141].
7. Efficient household water use strategies: Strategies such as low-flow appliances and public education about water conservation can significantly reduce domestic water use [142].
8. Construction of artificial reservoirs: Artificial reservoirs can store excess water during wet periods for use during dry periods, providing a buffer against climate variability.
9. River basin management: Integrated river basin management can ensure the equitable and sustainable use of water resources, considering both human needs and ecosystem health.
10. Restoration of natural water bodies: Natural water bodies, such as wetlands and rivers, can store water, buffer against extreme events, and provide important habitats. Their restoration can contribute to climate adaptation [143,144].

Briefly, water management strategies are integral to climate adaptation, ensuring the availability and quality of water resources in the face of changing climatic conditions.

3.4. Policy and Planning

Policies and planning are instrumental in adapting to climate change. They provide the institutional framework necessary for implementing both large-scale and local adaptation strategies, while also influencing individual behaviors and practices.

1. Incorporation of climate change into urban planning: By considering climate change in urban planning, cities can better prepare for future conditions, building resilience into their infrastructure and operations [145,146].
2. Development and enforcement of building codes related to climate change: Building codes that incorporate climate change risks can ensure that new construction is resilient to climate impacts, such as extreme heat, flooding, and storms [147,148].
3. Insurance policies reflecting climate risks: Insurance that accurately reflects climate risks can incentivize property owners to adopt adaptive measures and provide financial protection against climate impacts [149].
4. Zoning laws to prevent building in high-risk areas: Zoning can prevent the construction of new buildings in areas at high risk from climate impacts, such as floodplains or wildfire-prone areas [150,151].
5. Policies supporting renewable energy adoption: Renewable energy policies can help mitigate climate change and reduce the vulnerability of energy systems to climate impacts [152].
6. Tax incentives for climate-resilient practices: Tax incentives can encourage businesses and households to adopt practices that increase resilience, such as water conservation or the use of energy-efficient appliances [80].
7. Development of climate adaptation plans: Adaptation plans provide a roadmap for communities to build resilience, identifying key vulnerabilities and strategies to address them [153,154].
8. Education programs about climate change and adaptation strategies: Education can raise awareness about climate change and adaptation strategies, empowering individuals and communities to take action [154].
9. Climate migration policies: Some areas may become uninhabitable due to climate change, requiring policies to support the migration of affected communities [155].
10. Financial aid for affected communities: Financial aid can help communities recover from climate impacts and implement adaptation measures [156,157].

Lastly, policy and planning play vital roles in climate change adaptation, providing the regulatory and institutional context for effective action and ensuring that resources are available to those most in need.

3.5. Health and Social Measures

Health and social measures are a crucial aspect of climate adaptation, as climate change impacts are likely to pose significant risks to human health and societal structures. These measures primarily focus on preventing, reducing, or managing the negative health and societal effects associated with climate change.

1. Disease monitoring and prevention programs: Climate change can exacerbate certain diseases, especially those that are vector-borne, such as malaria and dengue. Disease monitoring and prevention programs can help predict outbreaks and reduce their impacts [158].
2. Heat-health action plans: Rising global temperatures can increase the risk of heat-related illnesses and deaths. Heat-health action plans, which include measures such as public awareness campaigns and early warning systems, can protect vulnerable populations during heatwaves [159].

3. Enhanced emergency services and infrastructure: The increased frequency and intensity of climate-related disasters necessitate enhanced emergency services and infrastructure to reduce disaster risks and manage disaster responses.
4. Food security programs: Climate change can threaten food security through impacts on crop yields and food prices. Programs aimed at enhancing food security can involve measures such as improved crop storage, crop insurance, and support for small-scale farmers [93,160].
5. Early warning systems for extreme weather events: Early warning systems can save lives and property by providing advance notice of extreme weather events, such as hurricanes, floods, and heatwaves [82,161].
6. Mental health support for climate trauma: The mental health impacts of climate change, including anxiety, depression, and post-traumatic stress disorder, are increasingly recognized. Providing mental health support is an important component of climate adaptation [162,163].
7. Community-based adaptation initiatives: These initiatives empower local communities to participate in the adaptation process, taking into account local knowledge and context to increase the effectiveness of adaptation strategies [164,165].
8. Provision of cooling centers for vulnerable populations: During extreme heat events, cooling centers can provide relief for those without access to air conditioning, particularly the elderly and low-income populations [166,167].
9. Training programs for new skills needed due to climate impacts: As climate change impacts various sectors, it may necessitate new skill sets. Training programs can help the workforce adapt to new demands [168,169].
10. Public health campaigns for climate-related diseases: Public health campaigns can raise awareness and provide information about the prevention and treatment of climate-related diseases [170].

Finally, health and social measures are essential for climate adaptation, as they focus on protecting human health and wellbeing and ensuring societal resilience in the face of climate change.

Table 2 presents a structured overview of 50 key measures for climate change adaptation, organized into five distinct categories. These categories reflect the various fields in which climate adaptation measures are applied: Physical Infrastructure, Ecosystem Restoration and Protection, Water Management, Policy and Planning, and Health and Social Measures. The outlined strategies span a wide array of approaches, from the reinforcement of physical infrastructure to better withstand climate-induced events to the development of public health campaigns specifically tailored for climate-related diseases.

Table 2. An overview of 50 climate change adaptation measures categorized by five major fields of application, each represented by a specific color chosen for its symbolic significance.

Physical Infrastructure	Ecosystem Restoration and Protection	Water Management	Policy and Planning	Health and Social Measures
Sea walls to protect against sea-level rise	Reforestation and afforestation	Rainwater harvesting systems	Incorporation of climate change into urban planning	Disease monitoring and prevention programs
Flood barriers and levees	Wetland restoration to absorb flood waters	Development of drought-tolerant crops	Development and enforcement of building codes related to climate change	Heat-health action plans
Drought-resistant agricultural practices	Coral reef restoration for coastal protection	Greywater recycling systems	Insurance policies reflecting climate risks	Enhanced emergency services and infrastructure
Upgrading stormwater and sewage systems	Protection and expansion of mangrove forests	Advanced irrigation techniques (e.g., drip irrigation)	Zoning laws to prevent building in high-risk areas	Food security programs

Table 2. Cont.

Physical Infrastructure	Ecosystem Restoration and Protection	Water Management	Policy and Planning	Health and Social Measures
Improving building insulation	Soil conservation practices	Desalination plants	Policies supporting renewable energy adoption	Early warning systems for extreme weather events
Heat-resistant pavement materials	Peatland restoration and conservation	Protection of groundwater resources	Tax incentives for climate-resilient practices	Mental health support for climate trauma
Construction of cooling centers	Establishment of wildlife corridors	Efficient household water use strategies	Development of climate adaptation plans	Community-based adaptation initiatives
Green roofs and walls	Planting urban trees for shade and cooling	Construction of artificial reservoirs	Education programs about climate change and adaptation strategies	Provision of cooling centers for vulnerable populations
Development of floating or stilt-based structures	Invasive species management	River basin management	Climate migration policies	Training programs for new skills needed due to climate impacts
Fire-resistant materials and designs for homes	Conservation of genetic diversity in crops and livestock	Restoration of natural water bodies	Financial aid for affected communities	Public health campaigns for climate-related diseases

In this representation, distinct colors have been thoughtfully assigned to each category to serve as intuitive markers, aiding in the swift comprehension of the overarching themes. These colors have been carefully chosen to encapsulate the essence of the respective fields of application, fostering a visual connection that deepens the understanding of the climate change adaptation measures delineated in Table 2.

- i. Physical Infrastructure (Purple): This category includes adaptation measures that focus on modifying, improving, or fortifying built structures and systems to withstand the impacts of climate change. The purple color signifies the tangible and visible nature of physical changes to infrastructure.
- ii. Ecosystem Restoration and Protection (Green): The green category encompasses adaptation measures that revolve around preserving, restoring, and enhancing natural ecosystems. The color green symbolizes the connection to nature and environmental well-being.
- iii. Water Management (Blue): Adaptation measures related to water management focus on strategies to mitigate the impacts of changing precipitation patterns, sea-level rise, and droughts. The blue color represents the critical role of water in climate adaptation efforts.
- iv. Policy and Planning (Black): The black category encompasses measures related to governance, regulations, and strategic planning. The black color symbolizes the authoritative and foundational nature of policy and planning.
- v. Health and Social Measures (Red): Adaptation measures focusing on health and social aspects address the potential impacts of climate change on human well-being. The red color signifies the urgency and potential risks associated with health and social challenges posed by climate change.

These color choices were made to visually reinforce the thematic focus of each category, enhancing the accessibility and interpretability of the information presented in the table.

These diverse climate adaptation measures emphasize the multidimensional nature of climate change and the need for comprehensive responses to mitigate its impacts. It is important to understand that the most effective adaptation strategies will often involve a combination of measures from multiple categories, depending on the specific regional climate vulnerabilities and available resources. By incorporating these measures in our planning and actions, we can be more resilient and adapt effectively to the changing climate.

conditions we face. Moreover, as the effects of climate change continue to evolve, it is critical to keep updating and expanding this list of measures to meet emerging challenges.

4. Application on Netobra.com: From Risk Detection to Adaptation

Netobra.com [170] is a GIS platform serving over 60,000 IMPIC-registered companies in the construction industry. In our study, we used Netobra.com to assess climate change risks across Portugal's 308 municipalities [171]. The platform revealed areas more vulnerable to climate change impacts and highlighted high-capacity companies in Lisbon and Porto as vital to the economy during climatic stress.

The platform also facilitates the integration of the 50 climate adaptation measures identified in our study into company profiles and project descriptions. These measures, spanning from physical infrastructure to policy and planning, help companies highlight their climate resilience strategies.

Overall, Netobra.com provides a valuable service for the construction industry, contributing to broader efforts to mitigate climate change and enhance urban resilience. It also assists policymakers in prioritizing resource allocation and risk management strategies (see Figures 4 and 5).

PT: Valor mediano de avaliacao bancaria (€-m²) - 2023 - Total
EN: Median bank appraisal value (€-m²) - 2023 - Total

Distrito

Aveiro

PORTUGAL	VALOR (VALUE) SUM		MEAN (AVERAGE)	
Grand Total	357,968.89		1,162.24	
CONCELHO	VALOR (VALUE)			
Águeda			968.33	
Albergaria-a-Velha			1003	
Anadia			888.67	
Arouca			1060.08	
Aveiro			1446	
Castelo de Paiva			942	
Espinho			1610	
Estarreja			1005.67	
Ílhavo			1352.67	
Mealhada			941	
Murtosa			1192.96	
Oliveira de Azeméis			947	
Oliveira do Bairro			952.33	
Ovar			1273	
Santa Maria da Feira			1068.67	
São João da Madeira			1020.33	
Sever do Vouga			868.17	
Vagos			1149.5	
Vale de Cambra			1136	
DISTRICTO	MEAN	SUM	MIN	MAX
Aveiro	1,096.07	20,825.38	868.17	1,610.00
Azores	1,032.53	19,618.11	678.70	1,387.45
Beja	953.28	13,345.92	684.05	1,771.31
Braga	1,082.91	15,160.75	861.22	1,360.33
Bragança	885.78	10,629.32	473.48	1,168.03
Castelo Branco	774.36	8,517.97	483.19	1,001.35
Coimbra	1,036.17	17,614.89	734.15	1,478.06
Évora	1,548.57	21,680.01	1,211.55	2,006.69
Faro	2,011.09	32,177.45	1,017.34	2,968.90
Guarda	776.47	10,870.53	596.68	981.55
Leiria	1,024.08	16,385.22	491.51	1,668.81
Lisboa	1,877.69	30,043.03	1,072.79	3,727.67
Madeira	1,459.80	16,057.82	981.21	1,932.42
Portalegre	831.40	12,471.01	610.06	992.82
Porto	1,301.92	23,434.49	866.51	2,580.67
Santarém	948.97	19,928.47	618.60	1,241.67
Setúbal	1,699.91	22,098.82	1,345.00	2,544.96
Viana do Castelo	1,029.93	10,299.32	793.97	1,326.91
Vila Real	976.93	13,677.05	685.18	1,329.47
Viseu	963.89	23,133.33	306.13	1,286.05

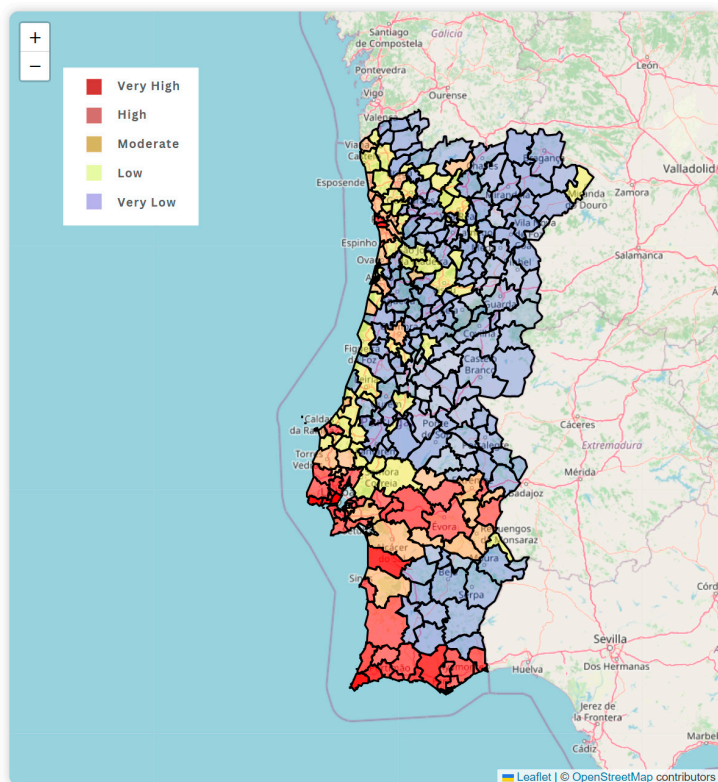


Figure 4. Interactive property value mapping on www.netobra.com (accessed on 22 August 2023) interactive map, indicating property values in Portugal's municipalities, which aids in understanding regional property variations for urban resilience [172].

PT: Índice composto de desenvolvimento regional (Índice geral) Anual 2019
EN: Regional Development Composite Index (General Index) Annual 2019

Distrito

Aveiro

PORTUGAL	VALOR (VALUE)	SUM	MEAN (AVERAGE)
Grand Total	29 738.61		96.55
CONCELHO	VALOR (VALUE)		
Águeda		101.71	
Albergaria-a-Velha		101.71	
Anadia		101.71	
Arouca		101.71	
Aveiro		101.71	
Castelo de Paiva		101.71	
Espinho		101.71	
Estarreja		101.71	
Ílhavo		101.71	
Mealhada		101.71	
Murtosa		101.71	
Oliveira de Azeméis		101.71	
Oliveira do Bairro		101.71	
Ovar		101.71	
Santa Maria da Feira		101.71	
São João da Madeira		101.71	
Sever do Vouga		101.71	
Vagos		101.71	
Vale de Cambra		101.71	

DISTRICTO	MEAN	SUM	MIN	MAX
Aveiro	101.71	1932.49	101.71	101.71
Azores	89.24	1695.56	89.24	89.24
Beja	94.84	1327.76	94.84	94.84
Braga	100.81	1411.30	97.79	101.63
Bragança	94.55	1134.60	94.55	94.55
Castelo Branco	92.51	1017.61	92.51	92.51
Coimbra	97.80	1662.60	97.80	97.80
Évora	95.48	1336.72	95.48	95.48
Faro	97.20	1555.20	97.20	97.20
Guarda	95.81	1341.39	88.71	96.36
Leiria	97.51	1560.16	97.51	97.51
Lisboa	106.37	1701.92	106.37	106.37
Madeira	97.29	1070.19	97.29	97.29
Portalegre	94.59	1418.85	94.59	94.59
Porto	102.89	1852.02	102.89	102.89
Santarém	94.40	1982.34	93.00	96.26
Setúbal	97.02	1261.26	97.02	97.02
Viana do Castelo	99.83	998.30	99.83	99.83
Vila Real	88.51	1239.14	88.36	88.71
Viseu	93.30	2239.20	88.71	95.19

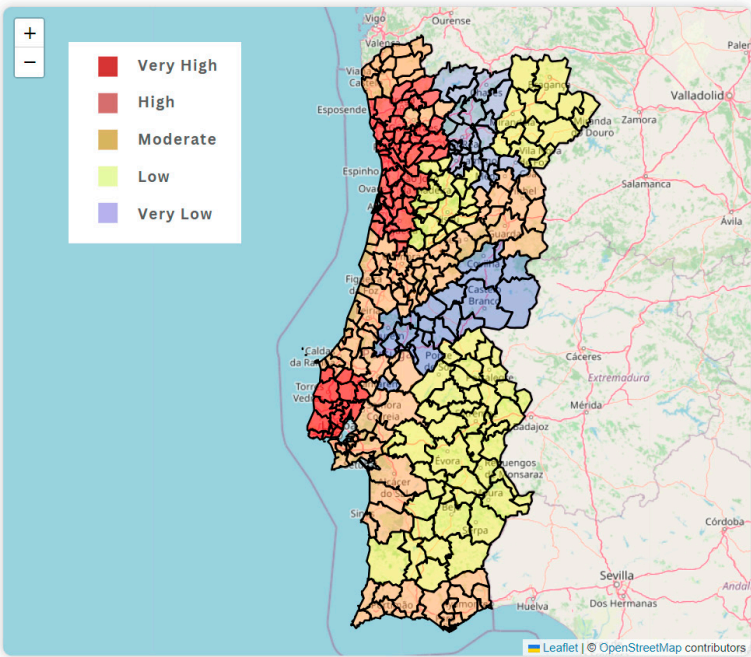


Figure 5. Regional Development Composite Index, Portugal’s municipal development index based on socio-economic indicators in www.netobra.com (accessed on 22 August 2023) [172].

4.1. The Role of Netobra.com in the Construction Industry

The construction industry is a key player in climate adaptation measures due to its role in urban development and infrastructure projects. By analyzing the number of certified companies per 1000 population across Portuguese districts, we used Netobra.com to assess regional economic robustness and industrial compliance. This analysis will help to target climate-adaptive resource distribution and risk management strategies, thus aiding the construction industry in building a resilient, sustainable, and robust economic framework for Portugal.

4.2. MCDA-AHP Mapping for Risk Detection on Netobra.com

The analytical hierarchy process (AHP) of multicriteria decision analysis (MCDA) is a structured technique for organizing and analyzing complex decisions. It involves breaking down a complex problem into its constituent parts (criteria and sub-criteria), evaluating these parts, and synthesizing the results to determine which options are the most viable.

In this study, we used the MCDA-AHP technique to create weighted maps that incorporate various geographic data based on subjective judgment [173]. The calculation step of MCDA-AHP involves first establishing a hierarchy of criteria for decision-making.

In the context of this study, these criteria might include factors such as population density, proximity to natural hazards, and infrastructure robustness (see Figure 6).

Each of these criteria is then compared pairwise to determine its relative importance, and a weight is assigned to each criterion based on this comparison. These weights are then used to calculate a score for each potential disaster hotspot on the map. The final map is created by overlaying these scores onto the geographic data.

These maps help identify potential disaster hotspots and critical assets that need prioritization in climate adaptation planning. The visualization of such data enables more informed, data-driven decision-making processes, thereby contributing significantly to urban resilience.

Weighting

Weighted Data Maps:

Select the maps that you want to weight and then weight them according to your opinion. The final decision will be a subjective decision that can reflect your opinion and create a new map on database.

Selecione os mapas que deseja ponderar e, em seguida, pondere-os de acordo com sua opinião. A decisão final será uma decisão subjetiva que pode refletir sua opinião e criar um novo mapa no banco de dados.

Select a Category

Todos

☐ Nome: População 25 - 64 anos-(Category: Geral)
 ☐ Nome: População 0 - 14 anos-(Category: Geral)
 ☐ Nome: Hospital Total-(Category: Saúde)
 ☐ Nome: Índice composto de desenvolvimento regional (Índice geral) Anual 2019-(Category: Economia)
 ☐ Nome: Floresta m2-(Category: Ambiente)
 ☐ Nome: Consumo de energia elétrica por habitante (kWh/ hab.) Anual-(Category: Habitação)
 ☒ Nome: Habitações convencionais - Anual-(Category: Habitação)
 ☐ Nome: Valor mediano das rendas por m2 de novos contratos de arrendamento de alojamentos familiares (€-2017-(Category: Economia)
 ☐ Nome: Valor mediano de avaliação bancária (€- m2) - 2023 - Total-(Category: Economia)
 ☐ Nome: Valor mediano de avaliação bancária (€- m2) - 2023 - Apartamentos-(Category: Economia)
 ☐ Nome: Valor mediano de avaliação bancária (€- m2) - 2023 - Moradias-(Category: Economia)
 ☐ Nome: Valor mediano de avaliação bancária (€- m2) - 2022 - Apartamentos-(Category: Economia)
 ☒ Nome: População 0-14 and 65+ anos-(Category: Geral)

median-value-of-rents-per-m2-of-new-lease-contracts-for-family-dwellings-(€)-2019-entry-39	4	median-value-of-rents-per-m2-of-new-lease-contracts-for-family-dwellings-(€)-2021-entry-69
median-value-of-rents-per-m2-of-new-lease-contracts-for-family-dwellings-(€)-2019-entry-39	3	population-0-14-and-65+-years-entry-42
median-value-of-rents-per-m2-of-new-lease-contracts-for-family-dwellings-(€)-2019-entry-39	1	conventional-housing---annual-entry-4
median-value-of-rents-per-m2-of-new-lease-contracts-for-family-dwellings-(€)-2021-entry-69	3	population-0-14-and-65+-years-entry-42
median-value-of-rents-per-m2-of-new-lease-contracts-for-family-dwellings-(€)-2021-entry-69	1/3	conventional-housing---annual-entry-4
population-0-14-and-65+-years-entry-42	1/3	conventional-housing---annual-entry-4

Calculate Weights

Map Name (PT):

Map Name (EN):

Map Description:

("median-value-of-rents-per-m2-of-new-lease-contracts-for-family-dwellings-(€)-2019-entry-39":0.3857,"median-value-of-rents-per-m2-of-new-lease-contracts-for-family-dwellings-(€)-2021-entry-69":0.1605,"population-0-14-and-65+-years-entry-42":0.0981,"conventional-housing---annual-entry-4":0.3557)

Map Category:

Todos

Create Map on Database

Calculated Matrix:

Consistency Ratio: -0.4775 %

The consistency ratio % is less than 0.5, you can submit your weights

MAP NAME	WEIGHT
median-value-of-rents-per-m2-of-new-lease-contracts-for-family-dwellings-(€)-2019-entry-39	38.57%
median-value-of-rents-per-m2-of-new-lease-contracts-for-family-dwellings-(€)-2021-entry-69	16.05%
population-0-14-and-65+-years-entry-42	9.81%
conventional-housing---annual-entry-4	35.57%

Figure 6. Disaster hotspots visualization that shows the use of MCDA-AHP mapping to identify high-risk climate impact areas, aiding urban resilience planning.

4.3. RIACT Process for Adaptation Measures on Netobra.com

Our application of the RIACT process, which includes Risk Identification, Analysis, Control, and Transfer, demonstrates the robust functionality of Netobra.com in the context

of climate adaptation measures. Through this process, we identified areas requiring more support, understood resource distribution needs, and managed risks related to climate crises or downturns. This comprehensive approach facilitated by Netobra.com underpins the successful implementation of climate adaptation measures, thus promoting urban resilience across Portuguese municipalities.

4.4. Evaluation of the Validity of the Netobra.com Platform

The role of Netobra.com, particularly in the phases of policy formulation and planning, is underscored as a pragmatic approach. It positions itself as a digital tool catering to decision-makers within the domain of climate adaptation and urban resilience.

The Netobra.com platform occupies a central position as an innovative digital solution within the forefront of climate adaptation and urban resilience initiatives. Its validation rests upon a robust foundation of data and steadfast methodologies, all while adhering to a commitment to accuracy and credibility that acknowledges the inherent complexities of the task at hand.

Netobra.com draws its validation from the meticulous integration of data sourced from the esteemed Instituto Nacional de Estatística (INE) in Portugal. This careful selection and utilization of authoritative data sources instills a sense of reliability and accentuate the platform's analytical credibility. By embracing established methodologies such as multicriteria decision analysis (MCDA) and the analytic hierarchy process (AHP) for hotspot identification, the platform not only strengthens its dependability but also promotes transparency throughout its operational procedures.

It is imperative to recognize that the current research serves as an initial demonstration of the platform's capabilities and limitations. Comprehensive validation necessitates subsequent phases of development and rigorous testing, as the platform remains in a state of continuous evolution. While the platform functions as an evaluative tool, its complete validation mandates real-world applications to truly ascertain its efficacy. This practical validation emphasizes the platform's practicality and pertinence within authentic climate adaptation scenarios. It is important to acknowledge that the timeframe required for a thorough validation cannot be hastened, as the platform's credibility necessitates time-intensive validation processes.

5. Results and Discussion

5.1. Interconnected Strategies for Climate Action: Leveraging Netobra.com's GIS Capabilities for Implementation and Analysis of Cascading Improvement Effect

Climate change impacts necessitate interconnected strategies that span multiple sectors. Our research reinforces the findings of previous studies, namely, that isolated efforts often fall short in effectively mitigating climate impacts. Our study, supported by Netobra.com's comprehensive GIS capabilities, undertakes an exploration of these interconnected strategies for climate action. It highlights how these strategies, spanning from physical infrastructure and climate adaptation measures to policy and planning initiatives, can create cascading improvement effects across diverse sectors.

Netobra.com aids the integration and analysis of the 50 climate adaptation measures identified in our study, reflecting prior research emphasizing the importance of comprehensive and multipronged approaches to climate adaptation. For instance, in the realm of physical infrastructure and climate adaptation measures, Netobra.com can provide spatial analysis and visualization of data related to measures such as sea walls, flood barriers, and heat-resistant pavement materials [174]. Similarly, in ecosystem restoration and protection efforts, Netobra.com can assist in mapping reforestation areas, wetland restoration sites, and wildlife corridors.

In the domain of water management, the platform can visualize measures such as rain-water harvesting systems, greywater recycling systems, and the construction of artificial reservoirs. From a policy and planning perspective, it can support efforts by mapping the incorporation of climate change into urban planning, zoning laws, and climate adaptation

plans. For health and social measures, Netobra.com can map and visualize disease monitoring and prevention programs, heat-health action plans, and community-based adaptation initiatives.

The integration of these measures within Netobra.com enhances decision-making, resource allocation, and monitoring processes, which ultimately contributes to the development of resilient and adaptive urban environments. Through a comprehensive understanding of these interrelationships, we can pave the way for holistic, multipronged climate action strategies. This exploration offers an in-depth analysis of these interconnections and their potential cascading effects.

In addressing climate change, it is vital to understand the interconnected nature of various efforts, and how they can create a cascading effect of improvements across multiple sectors. Table 3 illustrates this principle by categorizing various initiatives into five main categories: Physical Infrastructure and Climate Adaptation Measures, Ecosystem Restoration and Protection, Water Management, Policy and Planning, and Health and Social Measures. Each of these categories are then broken down into more specific measures. The last column of the table demonstrates the cascading improvement effects that can occur as a result of implementing these measures, as shown on Table 3.

Table 3. A summary of climate action strategies in five dimensions, showcasing their interconnected positive effects on urban resilience. To aid in the intuitive understanding of these dimensions and their synergies, a color code has been meticulously applied and explained in Table 2.

Main Category	Subcategory	Cascading Improvement Effects
Physical Infrastructure	Sea walls, flood barriers and levees, drought-resistant practices, upgrading stormwater and sewage systems, improving building insulation, heat-resistant pavement, construction of cooling centers, green roofs, floating structures, fire-resistant materials	Improved resilience to extreme weather, reduced heat effects, protected ecosystems, safeguarded communities, reduced infrastructure damage
Ecosystem Restoration and Protection	Reforestation, wetland restoration, coral reef restoration, mangrove forests expansion, soil conservation, peatland restoration, wildlife corridors, urban trees, invasive species management, conservation of genetic diversity	Enhanced biodiversity, better carbon sequestration, improved climate adaptation, protected wildlife habitats, improved soil quality and water absorption
Water Management	Rainwater harvesting, drought-tolerant crops, greywater recycling, advanced irrigation, desalination plants, groundwater resources, household water strategies, artificial reservoirs, river basin management, restoration of water bodies	Improved water availability, enhanced drought resilience, reduced water pollution, better agriculture yield, safeguarded aquatic ecosystems
Policy and Planning	Incorporation of climate change into urban planning, building codes, insurance policies, zoning laws, renewable energy policies, tax incentives, climate adaptation plans, education programs, climate migration policies, financial aid	Enhanced climate resilience, reduced risk, protected communities, improved living standards, better infrastructure, educated populace
Health and Social Measures	Disease monitoring, heat-health action plans, emergency services, food security programs, early warning systems, mental health support, community-based initiatives, cooling centers for vulnerable populations, training programs, public health campaigns	Improved public health, increased community resilience, reduced climate-induced mental health issues, safeguarded food supply, empowered communities, increased climate awareness

It is important to remember that these cascading effects are not isolated. For instance, implementing heat-health action plans not only contributes to improved public health, but also to increased community resilience, which in turn influences policy and planning. Similarly, practices such as reforestation and conservation of genetic diversity not only enhance biodiversity and carbon sequestration, but also provide a stronger base for physical

infrastructure and climate adaptation measures. It is this interconnected nature of climate action that makes holistic, cross-sectoral strategies so crucial for effective climate adaptation and mitigation. The above table offers a simple yet comprehensive depiction of this interconnectedness, shedding light on the potential for strategic multipronged climate action.

5.2. Interpretation and Implications for Urban Resilience and the Construction Industry

The integration of the 50 climate adaptation measures into Netobra.com has significant implications for urban resilience and the construction industry, highlighting the critical role of spatial planning and GIS capabilities in climate adaptation strategies. By leveraging the platform's GIS capabilities and mapping functionality, decision-makers can gain valuable insights into the distribution, effectiveness, and implementation status of these measures. This information can inform strategic planning, resource allocation, and policy development to enhance urban resilience in the face of climate change.

The comprehensive mapping of physical infrastructure measures, such as sea walls and flood barriers, and the upgrading of stormwater and sewage systems, allow for a better understanding of their spatial distribution and their contribution to mitigating the impacts of climate change. Decision-makers in the construction industry can utilize these insights to prioritize infrastructure projects, allocate resources effectively, and ensure that construction practices align with climate-resilient standards. This integration can ultimately enhance the industry's ability to deliver infrastructure that can withstand climate-related stresses and protect urban areas from potential hazards.

Analyzing and mapping restoration and protection measures such as reforestation and wetland restoration offer crucial insights into the spatial distribution of these efforts and their potential for urban resilience. This visualized data aids in identifying zones for ecological restoration, promoting biodiversity, and improving ecosystem capacity to counter climate threats. Such information is not only relevant for urban resilience but also vital for the construction industry, as it underscores the need for nature-based solutions in infrastructure projects.

Similarly, mapping of water management strategies, including rainwater harvesting and the protection of groundwater resources, elucidates their geographic distribution and their ability to bolster water security. This information guides decision-makers to identify areas needing such measures, optimize resource allocation, and encourage sustainable water practices. Furthermore, mapping policy measures such as climate change incorporation into urban planning and development of climate adaptation plans allows policymakers to visualize implementation, identify policy gaps, and strengthen climate-resilient planning frameworks. Lastly, mapping of health and social strategies such as disease monitoring programs and community-based initiatives helps highlight areas requiring interventions, enhancing community engagement in climate adaptation efforts. Each of these integrated facets underscores the need for a comprehensive approach to urban resilience, with the construction industry playing a pivotal role.

Overall, our study builds on previous research by demonstrating the practical implementation of a range of climate adaptation measures using Netobra.com GIS capabilities. The results offer valuable insights for enhancing urban resilience and promoting climate-resilient practices in the construction sector, contributing to the ongoing scholarly discourse on sustainable and adaptive urban development. By leveraging the platform's GIS capabilities, decision-makers can make informed decisions, allocate resources effectively, and promote climate-resilient practices in the construction sector. This integration represents a significant step forward in enhancing urban resilience and creating sustainable and adaptive cities.

5.3. Contribution to the Field and Research Limitations

This study contributes to the field of climate adaptation and urban resilience by demonstrating the practical application of Netobra.com in assessing and mapping climate

adaptation measures. The integration of these measures into a comprehensive platform enhances decision-making processes and promotes evidence-based planning and resource allocation. The study emphasizes the importance of considering multiple dimensions of urban resilience and provides valuable insights into the spatial distribution and potential effectiveness of various adaptation measures. This contribution supports policymakers, urban planners, and stakeholders in their efforts to build climate-resilient cities (see Figure 7).

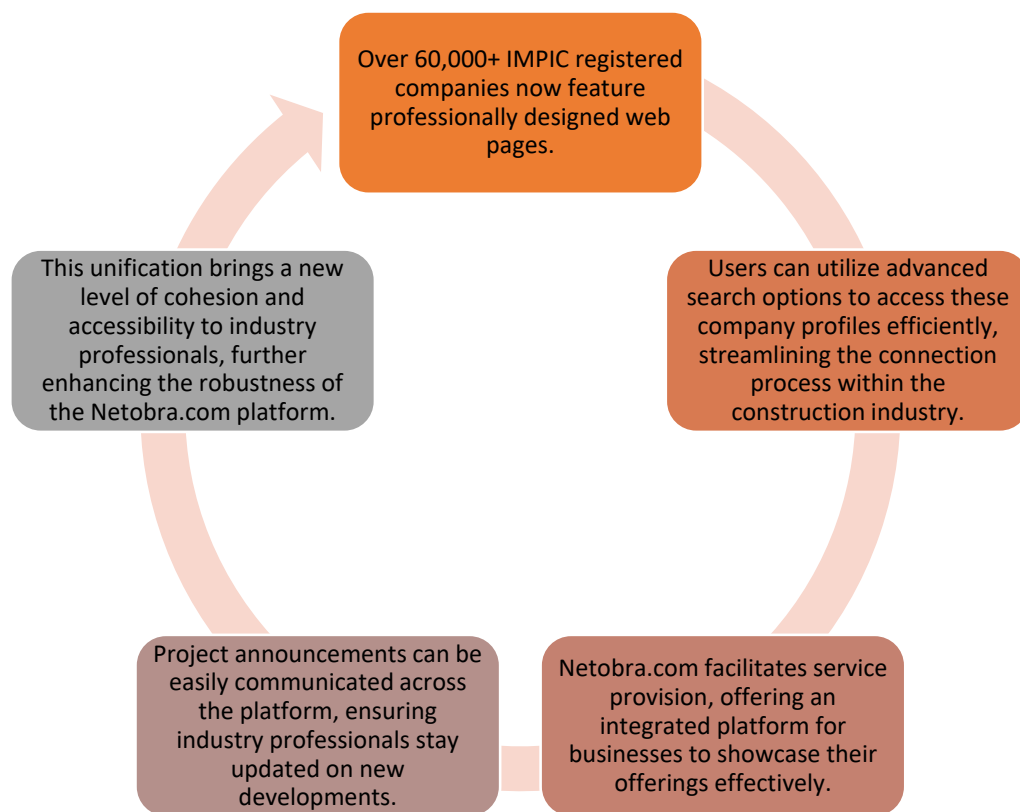


Figure 7. Key features of the Netobra.com platform, which highlights www.netobra.com (accessed on 22 August 2023) attributes—registered companies, search options, integrated services, and project announcements—enhancing industry connectivity and information sharing.

While this study provides valuable insights into climate adaptation measures for urban resilience, it is important to acknowledge its limitations. Firstly, the choice of Scopus as the primary database for the systematic review, while providing extensive coverage, may have resulted in the omission of relevant studies in other databases. Secondly, the MCDA-AHP method, while effective for prioritizing measures based on multiple criteria, relies on the subjective judgments of the decision-makers, which can introduce bias. Finally, the study’s focus on the Netobra.com platform, while providing a practical example of a digital tool for climate adaptation, limits the scope to other potentially useful digital tools.

6. Conclusions

The integration of the 50 climate adaptation measures into Netobra.com provides a powerful tool for enhancing urban resilience and promoting climate-resilient practices in the construction industry. The spatial mapping and analysis capabilities of the platform offer valuable insights into the distribution, effectiveness, and implementation status of these measures, enabling decision-makers to make informed decisions and allocate resources effectively. This integration emphasizes the importance of a holistic approach to urban resilience, considering physical infrastructure, ecosystem restoration, water management, policy and planning, and health and social measures. By incorporating these measures into

decision-making processes, cities can better prepare for the impacts of climate change and build a more resilient future.

6.1. Recap of the Study's Findings

Through the application of Netobra.com, this study provided a comprehensive assessment of the 50 climate adaptation measures across different categories. The analysis revealed the spatial distribution and potential impact of these measures on urban resilience. The findings highlighted the significance of physical infrastructure improvements, ecosystem restoration, water management strategies, policy and planning interventions, and health and social measures in enhancing urban resilience. The study underscored the importance of integrating these measures into decision-making processes to create more resilient and sustainable cities.

6.2. Suggestions for Future Research

While this study provides a comprehensive assessment of the 50 climate adaptation measures, there are several avenues for future research. First, further research could focus on the quantification and evaluation of the effectiveness of these measures in different urban contexts. This would provide more robust evidence of their impacts on urban resilience and inform decision-making processes. Second, research could explore the synergies and tradeoffs between different adaptation measures to optimize their implementation and maximize their benefits. Third, additional studies could investigate the financial and economic aspects of implementing these measures, including cost-effectiveness analyses and exploring funding mechanisms. Lastly, research could examine the social and equity dimensions of climate adaptation, ensuring that vulnerable communities are adequately considered in decision-making processes and adaptation strategies.

Overall, future research in this field will further enhance our understanding of climate adaptation measures and their integration into decision-making processes, leading to more effective and sustainable approaches to building urban resilience.

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Data Availability Statement: The dataset used in this research, which includes the data regarding the 50 climate adaptation measures and their distribution and potential impacts on urban resilience, is available at the following repository: NETOBRA—plataforma da indústria da construção. This data was used to perform the analysis and generate the findings reported in this study. For reasons of privacy and ethical restrictions, all personal identifiers have been removed from the dataset.

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References

- World Bank Disaster Risk Management Overview. Available online: <https://www.worldbank.org/en/topic/disasterriskmanagement/overview> (accessed on 17 August 2022).
- Catalano, M.; Forni, L.; Pezzolla, E. Climate-Change Adaptation: The Role of Fiscal Policy. *Resour. Energy Econ.* **2020**, *59*, 101111. [CrossRef]
- Beheshtian, A.; Donaghy, K.P.; Rouhani, O.M.; Geddes, R. Adaptation Planning for Climate-Resilient Urban Infrastructures. *Transp. Plan. Technol.* **2019**, *42*, 113–129. [CrossRef]
- Duarte, M.; Almeida, N.; Falcão, M.J.; Rezvani, S.M.H.S. *Resilience Rating System for Buildings against Natural Hazards*; Springer: Cham, Switzerland, 2022; ISBN 9783030967932.
- Garcia, J.; Rezvani, S.; Silva, M.J.F.; Almeida, N.; Pinto, C.; Gomes, R.; Ferreira, M.A.; Ribeiro, F.; Salvado, F.; Oliveira, C.S. *Resilience Assessment of Public Treasury Elementary School Buildings in Lisbon Municipality*; Springer: Cham, Switzerland, 2023; ISBN 9783031254475.
- ISO 31000 ISO 31000; Risk Management—Principles and Guidelines. International Organization for Standardization: Geneva, Switzerland, 2018; Volume 34.
- ISO 55000 ISO/CD: 2012; Asset Management—Overview Principles and Terminology. International Organization for Standardization: Geneva, Switzerland, 2014.
- The World Bank Building Urban Resilience*; World Bank Group: Washington, DC, USA, 2012.
- D’Agostino, A.L.; Sovacool, B.K. Sewing Climate-Resilient Seeds: Implementing Climate Change Adaptation Best Practices in Rural Cambodia. *Mitig. Adapt. Strateg. Glob. Chang.* **2011**, *16*, 699–720. [CrossRef]
- Engle, N.L.; De Bremond, A.; Malone, E.L. Towards a Resilience Indicator Framework for Making Climate-Change Adaptation Decisions. *Mitig. Adapt. Strateg. Glob. Chang.* **2013**, *19*, 1295–1312. [CrossRef]
- Osman, M.M.; Sevinc, H. Adaptation of Climate-Responsive Building Design Strategies and Resilience to Climate Change in the Hot/Arid Region of Khartoum, Sudan. *Sustain. Cities Soc.* **2019**, *47*, 101429. [CrossRef]
- Sethi, M.; Sharma, R.; Mohapatra, S.; Mittal, S. How to Tackle Complexity in Urban Climate Resilience? Negotiating Climate Science, Adaptation and Multi-Level Governance in India. *PLoS ONE* **2021**, *16*, e0253904. [CrossRef]
- Nguyen, Q.; Hoang, M.H.; Öborn, I.; van Noordwijk, M. Multipurpose Agroforestry as a Climate Change Resiliency Option for Farmers: An Example of Local Adaptation in Vietnam. *Clim. Chang.* **2013**, *117*, 241–257. [CrossRef]
- Lafond, V.; Lagarrigues, G.; Cordonnier, T.; Courbaud, B. Uneven-Aged Management Options to Promote Forest Resilience for Climate Change Adaptation: Effects of Group Selection and Harvesting Intensity. *Ann. For. Sci.* **2014**, *71*, 173–186. [CrossRef]
- Leandro, J.; Chen, K.-F.; Wood, R.R.R.; Ludwig, R. A Scalable Flood-Resilience-Index for Measuring Climate Change Adaptation: Munich City. *Water Res.* **2020**, *173*, 115502. [CrossRef] [PubMed]
- Kyriakopoulos, G.L.; Sebos, I. Enhancing Climate Neutrality and Resilience through Coordinated Climate Action: Review of the Synergies between Mitigation and Adaptation Actions. *Climate* **2023**, *11*, 105. [CrossRef]
- Devendiran, D.K.; Banerjee, S.; Mondal, A. Impact of Climate Change on Multihazard Performance of River-Crossing Bridges: Risk, Resilience, and Adaptation. *J. Perform. Constr. Facil.* **2021**, *35*, 04020127. [CrossRef]
- Brierley, G.; Fryirs, K. Truths of the Riverscape: Moving beyond Command-and-Control to Geomorphologically Informed Nature-Based River Management. *Geosci. Lett.* **2022**, *9*, 14. [CrossRef]
- Braunschweiler, D. Cross-Scale Collaboration for Adaptation to Climate Change: A Two-Mode Network Analysis of Bridging Actors in Switzerland. *Reg. Environ. Chang.* **2022**, *22*, 110. [CrossRef]
- Lamichhane, P.; Hadjikakou, M.; Miller, K.K.; Bryan, B.A. Climate Change Adaptation in Smallholder Agriculture: Adoption, Barriers, Determinants, and Policy Implications. *Mitig. Adapt. Strateg. Glob. Chang.* **2022**, *27*, 32. [CrossRef]
- Soumaoro, T. Adaptations of Market Garden Producers to Climate Change in Southern Mali. *GeoJournal* **2022**, *87*, 5413–5424. [CrossRef]
- Ruiz, I.; Pompeu, J.; Ruano, A.; Franco, P.; Balbi, S.; Sanz, M.J. Combined Artificial Intelligence, Sustainable Land Management, and Stakeholder Engagement for Integrated Landscape Management in Mediterranean Watersheds. *Environ. Sci. Policy* **2023**, *145*, 217–227. [CrossRef]
- Sibiya, N.; Sithole, M.; Mudau, L.; Simatele, M.D. Empowering the Voiceless: Securing the Participation of Marginalised Groups in Climate Change Governance in South Africa. *Sustainability* **2022**, *14*, 7111. [CrossRef]
- Bartelet, H.A.; Barnes, M.L.; Cumming, G.S. Determinants, Outcomes, and Feedbacks Associated with Microeconomic Adaptation to Climate Change. *Reg. Environ. Chang.* **2022**, *22*, 59. [CrossRef]
- Larson, A.J.; Jeronimo, S.M.A.; Hessburg, P.F.; Lutz, J.A.; Povak, N.A.; Cansler, C.A.; Kane, V.R.; Churchill, D.J. Tamm Review: Ecological Principles to Guide Post-Fire Forest Landscape Management in the Inland Pacific and Northern Rocky Mountain Regions. *For. Ecol. Manag.* **2022**, *504*, 119680. [CrossRef]
- Ikehi, M.E.; Ifeanyieze, F.O.; Onu, F.M.; Ejiofor, T.E.; Nwankwo, C.U. Assessing Climate Change Mitigation and Adaptation Strategies and Agricultural Innovation Systems in the Niger Delta. *GeoJournal* **2023**, *88*, 209–224. [CrossRef]
- Ricart, S.; Gandolfi, C.; Castelletti, A. Climate Change Awareness, Perceived Impacts, and Adaptation from Farmers’ Experience and Behavior: A Triple-Loop Review. *Reg. Environ. Chang.* **2023**, *23*, 82. [CrossRef]
- Loza, P.; Veloso-Gomes, F. Literature Review on Incorporating Climate Change Adaptation Measures in the Design of New Ports and Other Maritime Projects. *Sustainability* **2023**, *15*, 4569. [CrossRef]

29. Dill, H.G. A Geological and Mineralogical Review of Clay Mineral Deposits and Phyllosilicate Ore Guides in Central Europe—A Function of Geodynamics and Climate Change. *Ore Geol. Rev.* **2020**, *119*, 103304. [\[CrossRef\]](#)
30. Tsatsaris, A.; Kalogeropoulos, K.; Stathopoulos, N.; Louka, P.; Tsanakas, K.; Tsesmelis, D.E.; Krassanakis, V.; Petropoulos, G.P.; Pappas, V.; Chalkias, C. Geoinformation Technologies in Support of Environmental Hazards Monitoring under Climate Change: An Extensive Review. *ISPRS Int. J. Geoinf.* **2021**, *10*, 94. [\[CrossRef\]](#)
31. Williams, T.; Hardison, P. Culture, Law, Risk and Governance: Contexts of Traditional Knowledge in Climate Change Adaptation. *Clim. Chang.* **2013**, *120*, 531–544. [\[CrossRef\]](#)
32. Petzold, J.; Andrews, N.; Ford, J.D.; Hedemann, C.; Postigo, J.C. Indigenous Knowledge on Climate Change Adaptation: A Global Evidence Map of Academic Literature. *Environ. Res. Lett.* **2020**, *15*, 113007. [\[CrossRef\]](#)
33. Napogbong, L.A.; Domapielle, M.K.; Derbile, E.K. Indigenous Knowledge and Community-Based Risk Assessment of Climate Change among the Fulani Herder Community of Kpong, North-Western Ghana. *J. Water Clim. Chang.* **2021**, *12*, 484–501. [\[CrossRef\]](#)
34. Banda, V.D.; Dzwire, R.B.; Singh, S.K.; Kanyerere, T. Hydrological Modelling and Climate Adaptation under Changing Climate: A Review with a Focus in Sub-Saharan Africa. *Water* **2022**, *14*, 4031. [\[CrossRef\]](#)
35. Kim, H.; Marcouiller, D.W.; Woosnam, K.M. Multilevel Climate Governance, Anticipatory Adaptation, and the Vulnerability-Readiness Nexus. *Rev. Policy Res.* **2021**, *38*, 222–242. [\[CrossRef\]](#)
36. Nalau, J.; Cobb, G. The Strengths and Weaknesses of Future Visioning Approaches for Climate Change Adaptation: A Review. *Glob. Environ. Chang.* **2022**, *74*, 102527. [\[CrossRef\]](#)
37. Kabir Sarkar, M.S.; Begum, R.A.; Pereira, J.J.; Jaafar, A.H.B. Review of Costs and Methods for Climate Change Adaptation. *J. Environ. Sci. Technol.* **2012**, *5*, 396–406. [\[CrossRef\]](#)
38. Kourtis, I.M.; Tsihrintzis, V.A. Adaptation of Urban Drainage Networks to Climate Change: A Review. *Sci. Total Environ.* **2021**, *771*, 145431. [\[CrossRef\]](#) [\[PubMed\]](#)
39. Saraiva, A.; Monteiro, A. Climate Change as a Risk to Human Security: A Systematic Literature Review Focusing on Vulnerable Countries of Africa—Causes and Adaptation Strategies. *Int. J. Glob. Warm.* **2023**, *29*, 362–382. [\[CrossRef\]](#)
40. Magesa, B.A.; Mohan, G.; Matsuda, H.; Melts, I.; Kefi, M.; Fukushi, K. Understanding the Farmers' Choices and Adoption of Adaptation Strategies, and Plans to Climate Change Impact in Africa: A Systematic Review. *Clim. Serv.* **2023**, *30*, 100362. [\[CrossRef\]](#)
41. Meerow, S.; Newell, J.P.; Stults, M.; Newell, J.P.; Stults, M. Defining Urban Resilience: A Review. *Landsc. Urban Plan.* **2016**, *147*, 38–49. [\[CrossRef\]](#)
42. Büyüközkan, G.; Ilıcak, Ö.; Feyzioğlu, O. A Review of Urban Resilience Literature. *Sustain. Cities Soc.* **2022**, *77*, 103579. [\[CrossRef\]](#)
43. Cariolet, J.-M.M.; Vuillet, M.; Diab, Y. Mapping Urban Resilience to Disasters—A Review. *Sustain. Cities Soc.* **2019**, *51*, 101746. [\[CrossRef\]](#)
44. Rezvani, S.M.H.S.; Falcão, M.J.; Komljenovic, D.; de Almeida, N.M. A Systematic Literature Review on Urban Resilience Enabled with Asset and Disaster Risk Management Approaches and GIS-Based Decision Support Tools. *Appl. Sci.* **2023**, *13*, 2223. [\[CrossRef\]](#)
45. The World Bank. *Cities and Climate Change: An Urgent Agenda*; The World Bank: Washington, DC, USA, 2010.
46. Kuwae, T.; Crooks, S. Linking Climate Change Mitigation and Adaptation through Coastal Green–Gray Infrastructure: A Perspective. *Coast. Eng. J.* **2021**, *63*, 188–199. [\[CrossRef\]](#)
47. Espada, R.J.J.; Apan, A.; McDougall, K. Vulnerability Assessment and Interdependency Analysis of Critical Infrastructures for Climate Adaptation and Flood Mitigation. *Int. J. Disaster Resil. Built Environ.* **2015**, *6*, 313–346. [\[CrossRef\]](#)
48. Wamsler, C.; Niven, L.; Beery, T.H.; Bramryd, T.; Ekelund, N.; Jönsson, K.I.; Osmani, A.; Palo, T.; Stålhammar, S. Operationalizing Ecosystem-Based Adaptation: Harnessing Ecosystem Services to Buffer Communities against Climate Change. *Ecol. Soc.* **2016**, *21*, 31. [\[CrossRef\]](#)
49. Donatti, C.I.; Harvey, C.A.; Hole, D.; Panfil, S.N.; Schurman, H. Indicators to Measure the Climate Change Adaptation Outcomes of Ecosystem-Based Adaptation. *Clim. Chang.* **2020**, *158*, 413–433. [\[CrossRef\]](#)
50. Subak, S. Climate Change Adaptation in the U.K. Water Industry: Managers' Perceptions of Past Variability and Future Scenarios. *Water Resour. Manag.* **2000**, *14*, 137–156. [\[CrossRef\]](#)
51. Henao Casas, J.D.; Fernández Escalante, E.; Calero Gil, R.; Ayuga, F. Managed Aquifer Recharge as a Low-Regret Measure for Climate Change Adaptation: Insights from Los Arenales, Spain. *Water* **2022**, *14*, 3703. [\[CrossRef\]](#)
52. Street, R.; Opitz-Stapleton, S.; Nadin, R.; Chu, C.; Baum, S.; Conway, D. *Climate Change Adaptation Planning to Policy: Critical Considerations and Challenges*; Routledge: Oxfordshire, UK, 2015; ISBN 9781317593768.
53. Busayo, E.T.; Kalumba, A.M. Coastal Climate Change Adaptation and Disaster Risk Reduction: A Review of Policy, Programme and Practice for Sustainable Planning Outcomes. *Sustainability* **2020**, *12*, 6450. [\[CrossRef\]](#)
54. Rezvani, S.M.; de Almeida, N.M.; Falcão, M.J.; Duarte, M. Enhancing Urban Resilience Evaluation Systems through Automated Rational and Consistent Decision-Making Simulations. *Sustain. Cities Soc.* **2022**, *78*, 103612. [\[CrossRef\]](#)
55. Dong, S.; Abolfathi, S.; Salaududin, M.; Tan, Z.H.; Pearson, J.M. Enhancing Climate Resilience of Vertical Seawall with Retrofitting—A Physical Modelling Study. *Appl. Ocean. Res.* **2020**, *103*, 102331. [\[CrossRef\]](#)
56. Ghasemi Tousi, E.; O'Brien, W.; Doulabian, S.; Shadmehri Toosi, A. Climate Changes Impact on Stormwater Infrastructure Design in Tucson Arizona. *Sustain. Cities Soc.* **2021**, *72*, 103014. [\[CrossRef\]](#)

57. Long, W.-J.; Lin, C.; Tan, X.-W.; Tao, J.-L.; Ye, T.-H.; Luo, Q.-L. Structural Applications of Thermal Insulation Alkali Activated Materials with Reduced Graphene Oxide. *Materials* **2020**, *13*, 1052. [\[CrossRef\]](#)
58. Lorencová, E.K.; Whitham, C.E.L.; Bašta, P.; Harmáčková, Z.V.; Štěpánek, P.; Zahradníček, P.; Farda, A.; Vačkář, D. Participatory Climate Change Impact Assessment in Three Czech Cities: The Case of Heatwaves. *Sustainability* **2018**, *10*, 1906. [\[CrossRef\]](#)
59. Howarth, C.; Brooks, K. Decision-Making and Building Resilience to Nexus Shocks Locally: Exploring Flooding and Heatwaves in the UK. *Sustainability* **2017**, *9*, 838. [\[CrossRef\]](#)
60. Nahvi, A.; Sadati, S.M.S.; Cetin, K.; Ceylan, H.; Sassani, A.; Kim, S. Towards Resilient Infrastructure Systems for Winter Weather Events: Integrated Stochastic Economic Evaluation of Electrically Conductive Heated Airfield Pavements. *Sustain. Cities Soc.* **2018**, *41*, 195–204. [\[CrossRef\]](#)
61. Gernay, T.; Selamet, S.; Tondini, N.; Khorasani, N.E. Urban Infrastructure Resilience to Fire Disaster: An Overview. *Procedia Eng.* **2016**, *161*, 1801–1805. [\[CrossRef\]](#)
62. Ganteaume, A.; Barbero, R.; Jappiot, M.; Maillé, E. Understanding Future Changes to Fires in Southern Europe and Their Impacts on the Wildland-Urban Interface. *J. Saf. Sci. Resil.* **2021**, *2*, 20–29. [\[CrossRef\]](#)
63. Santala, K.; Cardou, F.; Yemshanov, D.; Campioni, F.; Simpson, M.; Handa, I.T.; Ryser, P.; Aubin, I. Finding the Perfect Mix: An Applied Model That Integrates Multiple Ecosystem Functions When Designing Restoration Programs. *Ecol. Eng.* **2022**, *180*, 106646. [\[CrossRef\]](#)
64. Su, M.; Wu, J.; Feng, M. Responses of Urban Ecosystem Vulnerability and Restoration Assessment on Typhoon Disaster: A Case Study of Zhuhai City, China. *J. Sens.* **2022**, *2022*, 8344619. [\[CrossRef\]](#)
65. Li, L.; Cheshmehzangi, A.; Chan, F.K.S.; Ives, C.D. Mapping the Research Landscape of Nature-Based Solutions in Urbanism. *Sustainability* **2021**, *13*, 3876. [\[CrossRef\]](#)
66. Bush, J.; Doyon, A. Building Urban Resilience with Nature-Based Solutions: How Can Urban Planning Contribute? *Cities* **2019**, *95*, 102483. [\[CrossRef\]](#)
67. Hazarika, H. Use of Low Carbon and Low Cost (LC2) Materials in Climate Change Adaptation Measures. *Geotech. Eng.* **2017**, *48*, 45–52.
68. Douglas, I. The Challenge of Urban Poverty for the Use of Green Infrastructure on Floodplains and Wetlands to Reduce Flood Impacts in Intertropical Africa. *Landsc. Urban Plan.* **2018**, *180*, 262–272. [\[CrossRef\]](#)
69. Saxena, A.; Buettner, W.C.; Kestler, L.; Kim, Y.-S. Opportunities and Barriers for Wood-Based Infrastructure in Urban Himalayas: A Review of Selected National Policies of Nepal. *Trees For. People* **2022**, *8*, 100244. [\[CrossRef\]](#)
70. Acosta, F.; Haroon, S. Memorial Parking Trees: Resilient Modular Design with Nature-Based Solutions in Vulnerable Urban Areas. *Land* **2021**, *10*, 298. [\[CrossRef\]](#)
71. *Genetically Engineered Crops: Experiences and Prospects*; National Academy of Sciences: Washington, DC, USA, 2017; ISBN 9780309437387.
72. Busscher, T.; van den Brink, M.; Verweij, S. Strategies for Integrating Water Management and Spatial Planning: Organising for Spatial Quality in the Dutch “Room for the River” Program. *J. Flood Risk Manag.* **2019**, *12*, e12448. [\[CrossRef\]](#)
73. Dallman, S.; Chaudhry, A.M.; Muleta, M.K.; Lee, J. Is Rainwater Harvesting Worthwhile? A Benefit-Cost Analysis. *J. Water Resour. Plan Manag.* **2021**, *147*, 04021011. [\[CrossRef\]](#)
74. Szpak, A.; Modrzyńska, J.; Piechowiak, J. Resilience of Polish Cities and Their Rainwater Management Policies. *Urban Clim.* **2022**, *44*, 101228. [\[CrossRef\]](#)
75. Hermanus, L.; Andrew, S. Community-Centred Infrastructure Design Process for Resilience Building in South African Informal Settlements: The “Genius of Space” Solid Waste and Greywater Infrastructure Project. *Case Stud. Environ.* **2018**, *2*, 1–10. [\[CrossRef\]](#)
76. Karrasch, L.; Siebenhüner, B.; Seibert, S.L. Groundwater Salinization in Northwestern Germany: A Case of Anticipatory Governance in the Field of Climate Adaptation? *Earth Syst. Gov.* **2023**, *17*, 100179. [\[CrossRef\]](#)
77. Aslam, R.A.; Shrestha, S.; Usman, M.N.; Khan, S.N.; Ali, S.; Sharif, M.S.; Sarwar, M.W.; Saddique, N.; Sarwar, A.; Ali, M.U.; et al. Integrated SWAT-MODFLOW Modeling-Based Groundwater Adaptation Policy Guidelines for Lahore, Pakistan under Projected Climate Change, and Human Development Scenarios. *Atmosphere* **2022**, *13*, 2001. [\[CrossRef\]](#)
78. Juhola, S.; Heikkinen, M.; Pietilä, T.; Groundstroem, F.; Käyhkö, J. Connecting Climate Justice and Adaptation Planning: An Adaptation Justice Index. *Environ. Sci. Policy* **2022**, *136*, 609–619. [\[CrossRef\]](#)
79. Allam, Z. Revisiting Energy Policy and Planning in Future Living Cities. *Biotechnol. Future Cities* **2020**, *35*, 35–52.
80. Gorji, M.-A.; Hadavi, N.; Nahavandi, N. Improving Urban Resilience against Climate Change through Government Tax Policies to Housing Companies: A Game-Theoretic Approach. *Urban Clim.* **2023**, *49*, 101565. [\[CrossRef\]](#)
81. Bruno, M.S. Resilience Engineering: A Report on the Needs of the Stakeholder Communities and the Prospects for Responsive Educational Programs. In Proceedings of the 2015 International Conference on Interactive Collaborative Learning, ICL 2015, Firenze, Italy, 20–24 September 2015; pp. 699–702.
82. Alcántara-Ayala, I.; Oliver-Smith, A. *The Necessity of Early Warning Articulated Systems (EWASs): Critical Issues beyond Response*; Springer: Berlin/Heidelberg, Germany, 2016; ISBN 9783319338804.
83. Bowen, K.J.; Ebi, K.; Friel, S.; McMichael, A.J. A Multi-Layered Governance Framework for Incorporating Social Science Insights into Adapting to the Health Impacts of Climate Change. *Glob. Health Action* **2013**, *6*, 21820. [\[CrossRef\]](#)
84. Kataria, I.; Morris, A.M.J. Addressing Chronic Noncommunicable Diseases Is Essential to Strengthen Urban Resilience to Communicable Pandemic Diseases. *Cities Health* **2020**, *5*, S97–S99. [\[CrossRef\]](#)

85. Hecht, A.A.; Biehl, E.; Barnett, D.J.; Neff, R.A. Urban Food Supply Chain Resilience for Crises Threatening Food Security: A Qualitative Study. *J. Acad. Nutr. Diet* **2019**, *119*, 211–224. [\[CrossRef\]](#)
86. Gasparini, P.; Manfredi, G.; Zschau, J. Earthquake Early Warning as a Tool for Improving Society's Resilience and Crisis Response. *Soil Dyn. Earthq. Eng.* **2011**, *31*, 267–270. [\[CrossRef\]](#)
87. Salata, F.; Golasi, I.; Petitti, D.; de Lieto Vollaro, E.; Coppi, M.; de Lieto Vollaro, A. *Relating Microclimate, Human Thermal Comfort and Health during Heat Waves: An Analysis of Heat Island Mitigation Strategies through a Case Study in an Urban Outdoor Environment*; Elsevier: Amsterdam, The Netherlands, 2017; Volume 30, ISBN 3906488012.
88. Chiabai, A.; Quiroga, S.; Martinez-Juarez, P.; Higgins, S.; Taylor, T. The Nexus between Climate Change, Ecosystem Services and Human Health: Towards a Conceptual Framework. *Sci. Total Environ.* **2018**, *635*, 1191–1204. [\[CrossRef\]](#)
89. Orimoloye, I.R.; Mazinyo, S.P.; Kalumba, A.M.; Ekundayo, O.Y.; Nel, W. Implications of Climate Variability and Change on Urban and Human Health: A Review. *Cities* **2019**, *91*, 213–223. [\[CrossRef\]](#)
90. Tan Phong, N.; Hao Quang, N.; Van Sang, T. Shoreline Change and Community-Based Climate Change Adaptation: Lessons Learnt from Brebes Regency, Indonesia. *Ocean Coast. Manag.* **2022**, *218*, 106037. [\[CrossRef\]](#)
91. Ozariso, B. Energy Effectiveness of Passive Cooling Design Strategies to Reduce the Impact of Long-Term Heatwaves on Occupants' Thermal Comfort in Europe: Climate Change and Mitigation. *J. Clean. Prod.* **2022**, *330*, 129675. [\[CrossRef\]](#)
92. Sharifi, A.; Pathak, M.; Joshi, C.; He, B.-J. A Systematic Review of the Health Co-Benefits of Urban Climate Change Adaptation. *Sustain. Cities Soc.* **2021**, *74*, 103190. [\[CrossRef\]](#)
93. Savage, A.; Schubert, L.; Huber, C.; Bambrick, H.; Hall, N.; Bellotti, B. Adaptation to the Climate Crisis: Opportunities for Food and Nutrition Security and Health in a Pacific Small Island State. *Weather. Clim. Soc.* **2020**, *12*, 745–758. [\[CrossRef\]](#)
94. Sahay, S. Adaptation to Health Outcomes of Climate Change and Variability at the City Level: An Empirical Decision Support Tool. *Sustain. Cities Soc.* **2019**, *47*, 101512. [\[CrossRef\]](#)
95. Mahany, M.J.; Keim, M.E. Challenges and Strategies for Climate Change Adaptation among Pacific Island Nations. *Disaster Med. Public Health Prep.* **2012**, *6*, 415–423. [\[CrossRef\]](#)
96. Jones, K.; Desai, A.; Brosnan, N.; Cooper, J.; Ali, F. Built Asset Management Climate Change Adaptation Model. *Int. J. Disaster Resil. Built Environ.* **2017**, *8*, 263–274. [\[CrossRef\]](#)
97. Rezvani, S.M.H.S.; Almeida, N.; Silva, M.J.F. *Multi-Disciplinary and Dynamic Urban Resilience Assessment through Stochastic Analysis of a Virtual City*; Springer: Cham, Switzerland, 2023; ISBN 9783031254475.
98. Sousa, V.; De Almeida, N.M.; Dias, L.A. Risk Management Framework for the Construction Industry According to the ISO 31000: 2009 Standard. *J. Risk Anal. Crisis Response* **2012**, *2*, 261–274. [\[CrossRef\]](#)
99. Almeida, N.M.; Silva, M.J.F.M.J.F.; Salvado, F.; Rodrigues, H.; Maletić, D.; de Almeida, N.M.N.M.; Silva, M.J.F.M.J.F.; Salvado, F.; Rodrigues, H.; Maletić, D. Risk-Informed Performance-Based Metrics for Evaluating the Structural Safety and Serviceability of Constructed Assets against Natural Disasters. *Sustainability* **2021**, *13*, 5925. [\[CrossRef\]](#)
100. Rasoulkhani, K.; Mostafavi, A.; Cole, J.; Sharvelle, S. Resilience-Based Infrastructure Planning and Asset Management: Study of Dual and Singular Water Distribution Infrastructure Performance Using a Simulation Approach. *Sustain. Cities Soc.* **2019**, *48*, 101577. [\[CrossRef\]](#)
101. Maletić, D.; Marques de Almeida, N.; Gomišček, B.; Maletić, M. Understanding Motives for and Barriers to Implementing Asset Management System: An Empirical Study for Engineered Physical Assets. *Prod. Plan. Control* **2022**, *33*, 1–17. [\[CrossRef\]](#)
102. Duarte, M.; De Almeida, N.M.; Falcão, M.J.; Salvado, F. Resilience of Constructed Assets against Natural Extreme Events from the Engineering Standpoint. In Proceedings of the International Conference of Construction, ENERGY, environment & Sustainability 2021, Coimbra, Portugal, 12–15 October 2021; pp. 1–8.
103. Espada, R.; Apan, A.; McDougall, K. Vulnerability Assessment of Urban Community and Critical Infrastructures for Integrated Flood Risk Management and Climate Adaptation Strategies. *Int. J. Disaster Resil. Built Environ.* **2017**, *8*, 375–411. [\[CrossRef\]](#)
104. Kim, Y.; Chester, M.V.; Eisenberg, D.A.; Redman, C.L. The Infrastructure Trolley Problem: Positioning Safe-to-Fail Infrastructure for Climate Change Adaptation. *Earths Future* **2019**, *7*, 704–717. [\[CrossRef\]](#)
105. Betzold, C.; Mohamed, I. Seawalls as a Response to Coastal Erosion and Flooding: A Case Study from Grande Comore, Comoros (West Indian Ocean). *Reg. Environ. Chang.* **2017**, *17*, 1077–1087. [\[CrossRef\]](#)
106. Oukes, C.; Leendertse, W.; Arts, J. Enhancing the Use of Flood Resilient Spatial Planning in Dutch Water Management. A Study of Barriers and Opportunities in Practice. *Plan. Theory Pract.* **2022**, *23*, 212–232. [\[CrossRef\]](#)
107. Burch, S.; Sheppard, S.R.J.; Shaw, A.; Flanders, D. Planning for Climate Change in a Flood-Prone Community: Municipal Barriers to Policy Action and the Use of Visualizations as Decision-Support Tools. *J. Flood Risk Manag.* **2010**, *3*, 126–139. [\[CrossRef\]](#)
108. Freire, P.; Rodrigues, M.; Fortunato, A.B.; Freitas, A. Flood and Drought Risk Assessment for Agricultural Areas (Tagus Estuary, Portugal). *Nat. Hazards Earth Syst. Sci.* **2021**, *21*, 2503–2521. [\[CrossRef\]](#)
109. Rahmati, O.; Falah, F.; Dayal, K.S.; Deo, R.C.; Mohammadi, F.; Biggs, T.; Moghaddam, D.D.; Naghibi, S.A.; Bui, D.T. Machine Learning Approaches for Spatial Modeling of Agricultural Droughts in the South-East Region of Queensland Australia. *Sci. Total Environ.* **2020**, *699*, 134230. [\[CrossRef\]](#)
110. Qiao, H.; Pei, J. Urban Stormwater Resilience Assessment Method Based on Cloud Model and Topsis. *Int. J. Environ. Res. Public Health* **2022**, *19*, 38. [\[CrossRef\]](#) [\[PubMed\]](#)
111. Li, A.; Han, H.; Yu, C. Secrecy Energy-Efficient UAV Communication via Trajectory Design and Power Control. *Wirel. Commun. Mob. Comput.* **2021**, *2021*, 9969311. [\[CrossRef\]](#)

112. Liu, L.; Wang, C.; Liang, Q. Preparation of a Heat Insulation Bonding Layer for Roads and Its Heat Insulation Effect. *J. Clean. Prod.* **2022**, *365*, 132828. [\[CrossRef\]](#)
113. He, W.; Zhang, J.; Li, H.; Liu, S.; Wang, Y.; Lv, B.; Wei, J. Optimal Thermal Management of Server Cooling System Based Cooling Tower under Different Ambient Temperatures. *Appl. Therm. Eng.* **2022**, *207*, 118176. [\[CrossRef\]](#)
114. Soltani, M.; Kashkooli, F.M.; Dehghani-Sanij, A.R.; Kazemi, A.R.; Bordbar, N.; Farshchi, M.J.; Elmi, M.; Gharali, K.; Dusseault, M.B. A Comprehensive Study of Geothermal Heating and Cooling Systems. *Sustain. Cities Soc.* **2019**, *44*, 793–818. [\[CrossRef\]](#)
115. Cappello, C.; Giuffrida, S.; Trovato, M.R.; Ventura, V. Environmental Identities and the Sustainable City. The Green Roof Prospect for the Ecological Transition. *Sustainability* **2022**, *14*, 2005. [\[CrossRef\]](#)
116. Zhang, G.; He, B.J.; Dewancker, B.J. The Maintenance of Prefabricated Green Roofs for Preserving Cooling Performance: A Field Measurement in the Subtropical City of Hangzhou, China. *Sustain. Cities Soc.* **2020**, *61*, 102314. [\[CrossRef\]](#)
117. Storbjörk, S.; Hjerpe, M. Stuck in Experimentation: Exploring Practical Experiences and Challenges of Using Floating Housing to Climate-Proof Waterfront Urban Development in Sweden. *J. Hous. Built Environ.* **2022**, *37*, 2263–2284. [\[CrossRef\]](#)
118. Adjaoud, A.; Puchot, L.; Federico, C.E.; Das, R.; Verge, P. Lignin-Based Benzoxazines: A Tunable Key-Precursor for the Design of Hydrophobic Coatings, Fire Resistant Materials and Catalyst-Free Vitrimers. *Chem. Eng. J.* **2023**, *453*, 139895. [\[CrossRef\]](#)
119. Hýsek, Š.; Frydrych, M.; Herclík, M.; Louda, P.; Fridrichová, L.; Le Van, S.; Le Chi, H. Fire-Resistant Sandwich-Structured Composite Material Based on Alternative Materials and Its Physical and Mechanical Properties. *Materials* **2019**, *12*, 1432. [\[CrossRef\]](#) [\[PubMed\]](#)
120. Dhillon, R.S.; von Wuehlisch, G. Mitigation of Global Warming through Renewable Biomass. *Biomass Bioenergy* **2013**, *48*, 75–89. [\[CrossRef\]](#)
121. Castillo-Uzcanga, M.M.; Barba-Macías, E.; Espinoza-Tenorio, A. *Freshwater Wetlands Conservation: An Assessment Protocol for Coastal Restoration in the Context of Climate Change*; Springer: Berlin/Heidelberg, Germany, 2018; ISBN 9783319905846.
122. Schernewski, G.; Schumacher, J.; Weisner, E.; Donges, L. A Combined Coastal Protection, Realignment and Wetland Restoration Scheme in the Southern Baltic: Planning Process, Public Information and Participation. *J. Coast. Conserv.* **2018**, *22*, 533–547. [\[CrossRef\]](#)
123. Beyer, H.L.; Kennedy, E.V.; Begger, M.; Chen, C.A.; Cinner, J.E.; Darling, E.S.; Eakin, C.M.; Gates, R.D.; Heron, S.F.; Knowlton, N.; et al. Risk-Sensitive Planning for Conserving Coral Reefs under Rapid Climate Change. *Conserv. Lett.* **2018**, *11*, e12587. [\[CrossRef\]](#)
124. Zhu, B.; Liao, J.; Shen, G. Spatio-Temporal Simulation of Mangrove Forests under Different Scenarios: A Case Study of Mangrove Protected Areas, Hainan Island, China. *Remote Sens.* **2021**, *13*, 4059. [\[CrossRef\]](#)
125. Servino, R.N.; Gomes, L.E.D.O.; Bernardino, A.F. Extreme Weather Impacts on Tropical Mangrove Forests in the Eastern Brazil Marine Ecoregion. *Sci. Total Environ.* **2018**, *628–629*, 233–240. [\[CrossRef\]](#)
126. Li, M.S.; Mao, L.J.; Shen, W.J.; Liu, S.Q.; Wei, A.S. Change and Fragmentation Trends of Zhanjiang Mangrove Forests in Southern China Using Multi-Temporal Landsat Imagery (1977–2010). *Estuar. Coast. Shelf Sci.* **2013**, *130*, 111–120. [\[CrossRef\]](#)
127. Mekonnen, M.; Getahun, M. Soil Conservation Practices Contribution in Trapping Sediment and Soil Organic Carbon, Minizir Watershed, Northwest Highlands of Ethiopia. *J. Soils Sediments* **2020**, *20*, 2484–2494. [\[CrossRef\]](#)
128. Harrison, M.E.; Ottay, J.B.; D’Arcy, L.J.; Cheyne, S.M.; Anggodo; Belcher, C.; Cole, L.; Dohong, A.; Ermiasi, Y.; Feldpausch, T.; et al. Tropical Forest and Peatland Conservation in Indonesia: Challenges and Directions. *People Nat.* **2020**, *2*, 4–28. [\[CrossRef\]](#)
129. Terzano, D.; Trezza, F.R.; Rezende, M.; Malatesta, L.; Lew Siew Yan, S.; Parish, F.; Moss, P.; Bresciani, F.; Cooke, R.; Dargusch, P.; et al. Prioritization of Peatland Restoration and Conservation Interventions in Sumatra, Kalimantan and Papua. *J. Nat. Conserv.* **2023**, *73*, 126388. [\[CrossRef\]](#)
130. Evans, J.P. Wildlife Corridors: An Urban Political Ecology. *Local Environ.* **2007**, *12*, 129–152. [\[CrossRef\]](#)
131. Tsoka, S.; Leduc, T.; Rodler, A. Assessing the Effects of Urban Street Trees on Building Cooling Energy Needs: The Role of Foliage Density and Planting Pattern. *Sustain. Cities Soc.* **2021**, *65*, 102633. [\[CrossRef\]](#)
132. Donovan, G.H.; Butry, D.T. The Value of Shade: Estimating the Effect of Urban Trees on Summertime Electricity Use. *Energy Build.* **2009**, *41*, 662–668. [\[CrossRef\]](#)
133. Havel, J.E.; Kovalenko, K.E.; Thomaz, S.M.; Amalfitano, S.; Kats, L.B. Aquatic Invasive Species: Challenges for the Future. *Hydrobiologia* **2015**, *750*, 147–170. [\[CrossRef\]](#)
134. Haider, H.; Ghuman, A.R.; Al-Salamah, I.S.; Ghazaw, Y.; Abdel-Maguid, R.H. Sustainability Evaluation of Rainwater Harvesting-Based Flood Risk Management Strategies: A Multilevel Decision-Making Framework for Arid Environments. *Arab. J. Sci. Eng.* **2019**, *44*, 8465–8488. [\[CrossRef\]](#)
135. Zhang, S.; Zhang, J.; Yue, T.; Jing, X. Impacts of Climate Change on Urban Rainwater Harvesting Systems. *Sci. Total Environ.* **2019**, *665*, 262–274. [\[CrossRef\]](#)
136. Mohsenipour, M.; Shahid, S.; Chung, E.-S.; Wang, X.-J. Changing Pattern of Droughts during Cropping Seasons of Bangladesh. *Water Resour. Manag.* **2018**, *32*, 1555–1568. [\[CrossRef\]](#)
137. Okada, M.; Iizumi, T.; Sakurai, G.; Hanasaki, N.; Sakai, T.; Okamoto, K.; Yokozawa, M. Modeling Irrigation-Based Climate Change Adaptation in Agriculture: Model Development and Evaluation in Northeast China. *J. Adv. Model. Earth Syst.* **2015**, *7*, 1409–1424. [\[CrossRef\]](#)
138. Haro-Monteaudo, D.; Palazón, L.; Zoumides, C.; Beguería, S. Optimal Implementation of Climate Change Adaptation Measures to Ensure Long-Term Sustainability on Large Irrigation Systems. *Water Resour. Manag.* **2023**, *37*, 2909–2924. [\[CrossRef\]](#)

139. Hadji, R.; Limani, Y.; Boumazbeur, A.E.; Demdoun, A.; Zighmi, K.; Zahri, F.; Chouabi, A. Climate Change and Its Influence on Shrinkage-Swelling Clays Susceptibility in a Semi-Arid Zone: A Case Study of Souk Ahras Municipality, NE-Algeria. *Desalination Water Treat.* **2014**, *52*, 2057–2072. [\[CrossRef\]](#)
140. Bhushan, R.; Ng, T.L. Integrating Desalination to Reservoir Operation to Increase Redundancy for More Secure Water Supply. *Water Resour. Res.* **2016**, *52*, 6137–6155. [\[CrossRef\]](#)
141. Jia, X.; O'Connor, D.; Hou, D.; Jin, Y.; Li, G.; Zheng, C.; Ok, Y.S.; Tsang, D.C.W.; Luo, J. Groundwater Depletion and Contamination: Spatial Distribution of Groundwater Resources Sustainability in China. *Sci. Total Environ.* **2019**, *672*, 551–562. [\[CrossRef\]](#)
142. Aihara, Y.; Shrestha, S.; Rajbhandari, S.; Bhattarai, A.P.; Bista, N.; Kazama, F.; Shindo, J. Resilience in Household Water Systems and Quality of Life after the Earthquake: A Mixed-Methods Study in Urban Nepal. *Water Policy* **2018**, *20*, 1013–1026. [\[CrossRef\]](#)
143. Mostafavi, A.; Ganapati, N.E.; Nazarnia, H.; Pradhananga, N.; Khanal, R. Adaptive Capacity under Chronic Stressors: Assessment of Water Infrastructure Resilience in 2015 Nepalese Earthquake Using a System Approach. *Nat. Hazards Rev.* **2018**, *19*, 05017006. [\[CrossRef\]](#)
144. Min, Z.Y.; Bhaktikul, K.; Aroonsrimorakot, S.; Sucharitakul, S.; Tabucanon, A.S.; Siswoyo, B.E. Factors Related to Coastal Communities' Water-Related Natural Disaster Awareness, Preparedness, Resilience and Recovery in Three Cyclone Nargis Affected Areas in the Ayeyarwaddy Delta Region, Myanmar. *Environ. Nat. Resour. J.* **2020**, *18*, 304–313. [\[CrossRef\]](#)
145. Hardoy, J.; Ruete, R. Incorporating Climate Change Adaptation into Planning for a Liveable City in Rosario, Argentina. *Environ. Urban* **2013**, *25*, 339–360. [\[CrossRef\]](#)
146. Kim, D.; Kang, J.E. Integrating Climate Change Adaptation into Community Planning Using a Participatory Process: The Case of Saebat Maeul Community in Busan, Korea. *Environ. Plan. B Urban Anal. City Sci.* **2018**, *45*, 669–690. [\[CrossRef\]](#)
147. Shapiro, S. The Realpolitik of Building Codes: Overcoming Practical Limitations to Climate Resilience. *Build. Res. Inf.* **2016**, *44*, 490–506. [\[CrossRef\]](#)
148. Nwadike, A.N.; Wilkinson, S. Why Amending Building Codes? An Investigation of the Benefits of Regular Building Code Amendment in New Zealand. *Int. J. Build. Pathol. Adapt.* **2022**, *40*, 76–100. [\[CrossRef\]](#)
149. Mendes-Da-Silva, W.; Flores, E.; Eckles, D.L. Informed Decisions Regarding Flood Events Induces Propensity for Insurances. *Environ. Sci. Policy* **2022**, *136*, 738–750. [\[CrossRef\]](#)
150. Räsänen, A.; Heikkinen, K.; Piila, N.; Juhola, S. Zoning and Weighting in Urban Heat Island Vulnerability and Risk Mapping in Helsinki, Finland. *Reg. Environ. Chang.* **2019**, *19*, 1481–1493. [\[CrossRef\]](#)
151. Wu, J.; Chen, Y.; Yang, R.; Zhao, Y. Exploring the Optimal Cost-Benefit Solution for a Low Impact Development Layout by Zoning, as Well as Considering the Inundation Duration and Inundation Depth. *Sustainability* **2020**, *12*, 4990. [\[CrossRef\]](#)
152. Pietrapertosa, F.; Khokhlov, V.; Salvia, M.; Cosmi, C. Climate Change Adaptation Policies and Plans: A Survey in 11 South East European Countries. *Renew. Sustain. Energy Rev.* **2018**, *81*, 3041–3050. [\[CrossRef\]](#)
153. Lysák, M.; Bugge-Henriksen, C. Current Status of Climate Change Adaptation Plans across the United States. *Mitig. Adapt. Strateg. Glob. Chang.* **2016**, *21*, 323–342. [\[CrossRef\]](#)
154. Mahlkow, N.; Donner, J. From Planning to Implementation? The Role of Climate Change Adaptation Plans to Tackle Heat Stress: A Case Study of Berlin, Germany. *J. Plan. Educ. Res.* **2017**, *37*, 385–396. [\[CrossRef\]](#)
155. Cassin, L.; Melindi-Ghidi, P.; Prieur, F. Confronting Climate Change: Adaptation vs. Migration in Small Island Developing States. *Resour. Energy Econ.* **2022**, *69*, 101301. [\[CrossRef\]](#)
156. Eise, J.; Lambert, N.J.; Wiemer, E.C. Leveraging Communities' Network Strengths to Support Climate Change Adaptation Information-Sharing: A Study with Coffee Farmers in Risaralda, Colombia. *Clim. Chang.* **2021**, *168*, 12. [\[CrossRef\]](#)
157. Pandey, R.; Alatalo, J.M.; Thapliyal, K.; Chauhan, S.; Archie, K.M.; Gupta, A.K.; Jha, S.K.; Kumar, M. Climate Change Vulnerability in Urban Slum Communities: Investigating Household Adaptation and Decision-Making Capacity in the Indian Himalaya. *Ecol. Indic.* **2018**, *90*, 379–391. [\[CrossRef\]](#)
158. Nichols, G.; Lake, I.; Heaviside, C. Climate Change and Water-Related Infectious Diseases. *Atmosphere* **2018**, *9*, 385. [\[CrossRef\]](#)
159. Petkova, E.P.; Bader, D.A.; Brooke Anderson, G.; Horton, R.M.; Knowlton, K.; Kinney, P.L. Heat-Related Mortality in a Warming Climate: Projections for 12 U.S. Cities. *Int. J. Environ. Res. Public Health* **2014**, *11*, 11371–11383. [\[CrossRef\]](#)
160. van Zonneveld, M.; Kindt, R.; McMullin, S.; Achigan-Dako, E.G.; N'Danikou, S.; Hsieh, W.-H.; Lin, Y.-R.; Dawson, I.K. Forgotten Food Crops in Sub-Saharan Africa for Healthy Diets in a Changing Climate. *Proc. Natl. Acad. Sci. USA* **2023**, *120*, e2205794120. [\[CrossRef\]](#)
161. Cuevas, S. Examining Climate Change Adaptation Measures: An Early Warning System in the Philippines. *Int. J. Clim. Chang. Strateg. Manag.* **2012**, *4*, 358–385. [\[CrossRef\]](#)
162. Wolf, T.; Lyne, K.; Martinez, G.S.; Kendrovski, V. The Health Effects of Climate Change in the WHO European Region. *Climate* **2015**, *3*, 901–936. [\[CrossRef\]](#)
163. Orsetti, E.; Tollin, N.; Lehmann, M.; Valderrama, V.A.; Morató, J. Building Resilient Cities: Climate Change and Health Interlinkages in the Planning of Public Spaces. *Int. J. Environ. Res. Public Health* **2022**, *19*, 1355. [\[CrossRef\]](#)
164. Dwirahmadi, F.; Rutherford, S.; Phung, D.; Chu, C. Understanding the Operational Concept of a Flood-Resilient Urban Community in Jakarta, Indonesia, from the Perspectives of Disaster Risk Reduction, Climate Change Adaptation and Development Agencies. *Int. J. Environ. Res. Public Health* **2019**, *16*, 3993. [\[CrossRef\]](#)

165. Tilstra, M.H.; Nielsen, C.C.; Tiwari, I.; Jones, C.A.; Vargas, A.O.; Quemerais, B.; Bulut, O.; Salma, J.; Yamamoto, S.S. Exploring Socio-Environmental Effects on Community Health in Edmonton, Canada to Understand Older Adult and Immigrant Risk in a Changing Climate. *Urban Clim.* **2022**, *44*, 101225. [CrossRef]
166. Sun, K.; Zhang, W.; Zeng, Z.; Levinson, R.; Wei, M.; Hong, T. Passive Cooling Designs to Improve Heat Resilience of Homes in Underserved and Vulnerable Communities. *Energy Build.* **2021**, *252*, 111383. [CrossRef]
167. Kim, Y.J.; Park, C.; Lee, D.K.; Park, T.Y. Connecting Public Health with Urban Planning: Allocating Walkable Cooling Shelters Considering Older People. *Landsc. Ecol. Eng.* **2023**, *19*, 257–269. [CrossRef]
168. Semien, J.; Nance, E. *Hazard Mitigation Training for Vulnerable Communities: A K.A.P.S. (Knowledge, Attitude, Preparedness, Skills) Approach*; Routledge: London, UK, 2022; ISBN 9781000573480.
169. Kim, K.; Wolshon, B.; Pant, P.; Yamashita, E.; Ghimire, J. Assessment of Evacuation Training Needs: Targeting Instruction to Meet the Requirements of Local Communities and Agencies. *J. Emerg. Manag.* **2020**, *18*, 475–487. [CrossRef]
170. Leal Filho, W.; Bönecke, J.; Spielmann, H.; Azeiteiro, U.M.; Alves, F.; Lopes de Carvalho, M.; Nagy, G.J. Climate Change and Health: An Analysis of Causal Relations on the Spread of Vector-Borne Diseases in Brazil. *J. Clean. Prod.* **2018**, *177*, 589–596. [CrossRef]
171. Rezvani, S.M.H.S.; Almeida, N.; Silva, M.J.F.; Maletič, D. *Resilience Exposure Assessment Using Multi-Layer Mapping of Portuguese 308 Cities and Communities*; Springer: Berlin/Heidelberg, Germany, 2023; ISBN 9783031254475.
172. Seyed Rezvani NETOBRA Web Application. Available online: <https://www.netobra.com/> (accessed on 30 June 2023).
173. Rezvani, S.; Almeida, N. Multi-Criteria Decision Analysis Applied to Subcontractors Selection in Construction Projects. In Proceedings of the 11th IMA International Conference on Modelling in Industrial Maintenance and Reliability, Virtual, 29 June–1 July 2021; Institute of Mathematics & Its Applications: Essex, UK, 2021.
174. Rezvani, S.; Gomes, M.C. Assessment of Pavement Degradation through Statistical Analysis Model A Case Study of the Department of Transportation (DOT) of Iowa, USA. In Proceedings of the 11th IMA International Conference on Modelling in Industrial Maintenance and Reliability, Virtual, 29 June–1 July 2021; Institute of Mathematics & Its Applications: Essex, UK, 2021.

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